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Structured Missingness in Synthetic Clinical Tabular Data: An Experimental Evaluation of Missingness Preservation, Predictive Performance, and Privacy

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ABSTRACT

Missing values are common in clinical tabular data and can appear in structured patterns. This thesis investigates missingness structure in the Chronic Kidney Disease (CKD) and Hepatitis datasets from the UCI Machine Learning Repository and investigates whether representing selected co-missingness patterns affects synthetic data generation. We identify co-missingness patterns in both datasets and evaluate whether missingness can be predicted from observed variables. We then compare two missingness representation strategies based on downstream predictive performance, missingness pattern preservation, predictive performance between groups, and privacy exposure. The results show stronger and more interpretable co-missingness structure in CKD than in Hepatitis. The two representation strategies give mixed results. In CKD, explicitly representing the identified co-missingness patterns gives better missingness pattern preservation, while in Hepatitis the other strategy performs better. Differences in downstream predictive performance are generally small. The subgroup results vary between subgroups and datasets, and the CKD outlier group shows higher privacy exposure. Overall, the results do not show a consistent advantage from explicitly representing co-missingness for synthetic data generation in the experiments performed in this thesis.

1 INTRODUCTION

Historical patient data collected within health care systems is of high value for clinical research. It supports the development, validation, and use of artificial intelligence (AI) and machine learning (ML) tools in clinical practice [46]. Realizing the full potential of these data, however, depends on the ability to share and reuse them across studies and institutions. Much of this data cannot be freely shared because patient records contain personally identifiable and sensitive information and are therefore subject to strict confidentiality requirements and regulatory constraints such as the European Union's General Data Protection Regulation (GDPR) [46, 52]. Before submitting a formal access request, researchers need to determine whether a dataset is suitable for their purposes. In practice, they are largely limited to browsing PDF codebooks. These are static documents describing a dataset's variables, codes, and structure, with search typically restricted to exact titles and keywords [30]. This makes it difficult to assess whether a dataset's variables, units, and coding schemes match the research question.

The Findability, Accessibility, Interoperability, and Reusability (FAIR) data principles provide a framework for improving the discovery and reuse of research data through machine-actionable metadata [50]. These are structured descriptions that can be interpreted by both humans and machines. They enable datasets to be more easily discovered, understood, and assessed for relevance without requiring direct access to the underlying data. For restricted clinical datasets, this creates the possibility of exposing richer information about the content of a dataset while preserving patient confidentiality [30].

Several metadata standards have been proposed to support FAIR data practices. Examples include CSV on the Web, RDF Data Cube, VoID, the Semantic Sensor Network ontology, and I-ADOPT. While these standards provide mechanisms for describing datasets and promoting interoperability, they do not necessarily capture the structural, statistical, and semantic characteristics of restricted clinical tabular data in sufficient detail for dataset discovery and assessment [30]. This leaves a need for metadata that can describe the characteristics of clinical tabular data in a way that is useful for researchers before they obtain access to the underlying data.

In cases where metadata enables researchers to identify a promising dataset, access requests can still take months. During this period, researchers are unable to inspect the data directly, making it difficult to understand its actual properties, validate assumptions, and determine whether the study requirements are met. Synthetic tabular data can provide a practical alternative for first-look exploration and method development during this waiting period [18, 25, 46]. Synthetic tabular data consists of artificially generated records that aim to preserve important properties and relationships found in the real data. This makes synthetic data potentially useful as a preview of restricted datasets. However, synthetic data is not automatically private, and its usefulness depends on which properties of the original data are preserved. Evaluating synthetic data therefore requires consideration of its statistical properties, downstream utility, privacy, and subgroup representation. It is not enough to rely only on overall similarity between real and synthetic data [24].

Privacy risk is also not necessarily distributed uniformly across the population. Patients with rare diseases or unusual characteristics may be more distinctive and therefore more vulnerable to disclosure. At the same time, reducing the representation of such groups in synthetic data may reduce its usefulness for clinical research. This creates a tension between privacy, utility, and subgroup representation [24, 46]. Information about these properties is therefore important when synthetic data is used as a preview of restricted clinical data.

Beyond privacy and fairness, clinical datasets present a specific challenge for both metadata descriptions and synthetic data generation in the form of missing data. Records are rarely complete

and, in some settings, 30–40% of variables are missing more than half of their expected values [19]. More importantly, missingness is often not random but structured and can result from clinical workflows and diagnostic decisions [39]. The mechanisms underlying missingness affect how missing values can be handled and how the resulting data should be interpreted. Properly describing clinical data therefore requires metadata that captures relevant missingness structure.

These challenges suggest that the way missingness is represented during synthetic data generation may affect how well the synthetic data preserve the original data. It may also affect downstream predictive performance, privacy exposure, and differences between patient groups. We therefore investigate whether explicitly representing variables that tend to be missing together affects these properties.

More specifically, we consider three aspects:

(1) We examine the missingness structure in two clinical tabular datasets and identify groups of variables that tend to be missing together. We represent these groups as structured metadata elements.
(2) We compare two missingness representation strategies during synthetic data generation and examine their effects on downstream predictive performance and preservation of missingness patterns.
(3) We compare the synthetic data across patient groups in terms of predictive performance and singling-out risk.

2 RESEARCH QUESTIONS

This thesis has three research questions. First, we characterize the missingness structure in clinical tabular data using two datasets. Second, we compare two ways of representing this structure during synthetic data generation and examine their effects on downstream predictive performance and missingness preservation. Finally, we compare the resulting synthetic data across subgroups in terms of predictive performance and privacy exposure.

RQ1: What missingness structure can be identified in clinical tabular data, and how predictable is missingness from the observed data?

- **RQ1.1:** What co-missingness patterns can be identified in clinical tabular data, and how can these patterns be represented as structured metadata elements?
- **RQ1.2:** To what extent can the missingness of a variable be predicted from the observed values of other variables, and what does this indicate about the MCAR assumption?

RQ2: How do different missingness representation strategies affect downstream predictive performance and missingness-pattern preservation in synthetic data?

- **RQ2.1:** How does each missingness strategy affect downstream predictive performance?
- **RQ2.2:** How closely does each missingness strategy reproduce the missingness-pattern distribution of the real data?

RQ3: How do the missingness representation strategies compare across subgroups in terms of predictive performance and privacy exposure?

- **RQ3.1:** What differences in predictive performance are observed between subgroups and the rest of the population in the real and synthetic data?
- **RQ3.2:** How does privacy exposure differ between subgroups and the rest of the population under the different missingness representation strategies?

3 BACKGROUND

3.1 Missing data mechanisms

The standard taxonomy of missing data mechanisms was introduced by Rubin [41] and later described in more detail by Little and Rubin [28]. The three mechanisms describe how missingness can depend on information in the data.

Data is *missing completely at random* (MCAR) when missingness does not depend on either observed or unobserved values. Data is *missing at random* (MAR) when missingness may depend on observed values, but does not depend on the missing values after conditioning on the observed data. Data is *missing not at random* (MNAR) when missingness depends on the unobserved values even after conditioning on the observed data. MCAR is a special case of MAR, so the two mechanisms are not mutually exclusive.

The distinction is important when deciding how to handle missing data. Under MCAR, the missingness does not provide additional information about the data. Under MAR, observed variables can provide information about the missingness and can be used when estimating missing values. Under MNAR, the missingness may depend on values that are not observed, so this information cannot be obtained from the observed data alone.

3.2 Synthetic data generation approaches

Synthetic tabular data can be generated using different approaches that differ in how they model relationships between variables and generate new observations [13]. Bayesian networks represent dependencies between variables using a directed acyclic graph. Each node represents a variable, and directed edges represent conditional dependencies between variables. The joint distribution can then be factorized according to the network structure [6].

Bayesian networks can be used to generate synthetic observations by sampling from the conditional distributions represented by the network. After learning the network structure and conditional distributions from the original data, new observations can be generated from these distributions. This allows dependencies between variables in the original data to be represented in the synthetic data.

3.3 Evaluation of synthetic data

Synthetic data can be evaluated from several perspectives because different properties of the original data may be preserved to different degrees. Statistical fidelity, downstream utility, privacy, and fairness provide different views of synthetic data quality [27].

Statistical fidelity measures how closely the statistical properties of the real data are reproduced in the synthetic data. This includes individual variable distributions, relationships between variables, and the overall structure of the data. *Downstream utility* measures whether the synthetic data retains information that is useful for a particular analytical or predictive task. A synthetic dataset can have similar statistical properties to the real data while still losing

information that is important for a specific task. *Privacy* concerns the possibility that the synthetic data reveals information about individuals in the original data. This can include information that allows individuals or sensitive characteristics to be identified, linked, or inferred. *Fairness* concerns whether synthetic data results in comparable outcomes across population subgroups. Different fairness measures capture different aspects of these differences.

These aspects can involve a trade-off. Preserving more information from the original data can improve utility but may also increase privacy risks, while stronger privacy protection can reduce the information retained in the synthetic data [16].

3.4 Train on Synthetic, Test on Real framework

Train on Synthetic, Test on Real (TSTR) evaluates whether synthetic data retains information that is useful for a downstream predictive task [12]. A predictive model is trained on synthetic training data and evaluated on real held-out data. Its performance is compared with a model trained on the real training data and evaluated on the same real held-out data. The evaluation data therefore remains fixed, while the training data differs between the real and synthetic conditions.

3.5 Multiple Imputation by Chained Equations (MICE)

Multiple Imputation by Chained Equations (MICE) is a method for imputing missing values in multivariate data [47]. Instead of imputing each variable independently, MICE models each incomplete variable using the other variables in the dataset. A separate model can be used for each variable depending on its type. The procedure iterates over the incomplete variables, fitting a model using the observed and currently imputed values and then updating the missing values. This process is repeated for several iterations.

MICE can generate multiple completed datasets by repeating the imputation process with different random draws. These datasets can then be analysed separately and combined using Rubin's rules, which accounts for uncertainty introduced by the missing values [33]. Compared with simple methods such as mean or median imputation, MICE can better preserve relationships between variables because each variable is imputed using information from the other variables included in the imputation model.

The standard MICE approach relies on assumptions under which the missing values can be estimated from the observed data. When missingness depends on unobserved values, standard MICE cannot directly account for this dependence using the observed data alone. MNAR settings therefore require additional assumptions or sensitivity analyses.

3.6 Clustering and outlier detection

Clustering is an unsupervised learning approach that groups observations according to their similarity across a set of features. k-means clustering partitions observations into k clusters based on their distance to a cluster centroid. The centroids are updated iteratively based on the observations assigned to each cluster [29].

The choice of k can affect both the cluster assignments and the interpretation of the resulting groups. Silhouette width measures how well an observation fits within its assigned cluster by comparing its average distance to observations in its own cluster with its average distance to observations in the nearest other cluster. Higher values indicate better separation between clusters [40]. Clustering stability can also be assessed by repeating the clustering with different random initializations. The adjusted Rand Index (ARI) measures the agreement between two cluster assignments while correcting for agreement expected by chance [21].

Clustering can also be used for outlier detection. The goal of outlier detection is to identify observations that differ from the rest of the data. Several approaches can be used, including statistical, distance-based, density-based, clustering-based, graph-based, ensemble-based, and learning-based methods [48]. In clustering-based outlier detection, the distance of an observation from its assigned cluster can indicate how unusual it is relative to that cluster.

Decision trees can then be used to characterize the resulting groups. They recursively split observations based on their feature values and produce interpretable rules that show which features distinguish the groups [31].

4 DESIGN

This section describes the experimental design used to answer the research questions. It covers the datasets, missingness analysis, missingness representation strategies, synthetic data generation, and evaluation methods.

4.1 Datasets

The framework is evaluated on two clinical datasets chosen to represent different domains, different missingness structures, and different dataset sizes.

4.1.1 Main dataset. The primary dataset is the Chronic Kidney Disease (CKD) dataset from the UCI Machine Learning Repository [42]. It has 400 patient records described in 25 attributes (11 numeric and 13 nominal) and a binary class label for the presence or absence of chronic kidney disease. Of the 400 records, only 158 contain complete information. The variables include results of tests used in the diagnosis and monitoring of CKD [22]. Because many of these tests are ordered selectively and depend on a patient's symptoms and disease stage, missing values may not be random and may reflect the clinical relevance behind them [22, 45]. This makes the dataset suitable for treating missingness as structured rather than random. Table 1 shows summarized statistics for continuous variables, stratified by CKD status. As expected, several variables indicate a separation between groups, e.g. serum creatinine (sc) is higher in CKD patients (median 2.3 vs 0.9 mg/dL), while hemoglobin (hemo) (median 10.9 vs 15 g/dL) is lower. More interestingly, missingness varies across variables, with values for blood pressure (bp) missing the least and tests that measure packed cell volume (pcv), white blood cell count (wbcc) and red blood cell count (rbcc) missing the most. This supports the claim that missingness depends on the tests that are ordered selectively. Table 2 shows summarized statistics for categorical variables, stratified by CKD status. As with the continuous variables, the categorical variables show a clear separation between CKD and non-CKD patients, both in their values and in their missingness.

Table 1: Main statistics of CKD dataset – continuous variables

Variable	CKD (N = 250)	Non-CKD (N = 150)	Miss.
age	59 (47, 66) [2, 90]	46 (34, 58) [12, 80]	9
bp	80 (70, 90) [50, 180]	70 (60, 80) [60, 80]	12
bgr	144 (107, 220) [22, 490]	108 (94, 124) [70, 140]	44
bu	53 (32, 93) [2, 391]	33 (24, 44) [10, 50]	19
sc	2.3 (1.4, 4.6) [0.5, 76.0]	0.9 (0.6, 1.1) [0.4, 1.2]	17
sod	136 (132, 139) [5, 163]	141 (138, 146) [135, 150]	87
pot	4.30 (3.80, 4.90) [2.50, 47.00]	4.50 (3.70, 4.90) [3.30, 5.00]	88
hemo	10.90 (9.45, 12.05) [3.10, 16.10]	15.00 (14.10, 16.20) [13.00, 17.80]	52
pcv	33 (28, 37) [9, 52]	46 (43, 50) [40, 54]	71
wbcc	8,800 (6,700, 10,700) [2,200, 26,400]	7,500 (6,300, 9,300) [4,300, 11,000]	106
rbcc	3.90 (3.40, 4.40) [2.10, 8.00]	5.30 (4.90, 5.90) [4.40, 6.50]	131

Note: Median (Q1, Q3); [Min, Max] shown below. Miss. = number of missing values (total across all patients).
Abbreviations: age = age (years); bp = blood pressure (mm Hg); bgr = blood glucose (mg/dL); bu = blood urea (mg/dL); sc = serum creatinine (mg/dL); sod = sodium (mEq/L); pot = potassium (mEq/L); hemo = hemoglobin (g/dL); pcv = packed cell volume; wbcc = WBC count (cells/cumm); rbcc = RBC count (millions/cmm).

Table 2: Main statistics of CKD dataset – categorical variables

Variable	CKD (N = 250)		Non-CKD (N = 150)		Miss.
sg	1.005: 3.4%, 1.010: 40% 1.015: 36%, 1.020: 15% 1.025: 5.3%		1.005–1.015: 0% 1.020: 52%, 1.025: 48%		47
al	0: 26%, 1: 21%, 2: 21% 3: 21%, 4: 11%, 5: 0.5%		0: 100% 1–5: 0%		46
su	0: 70%, 1: 6.3%, 2: 8.7% 3: 6.8%, 4: 6.3%, 5: 1.5%		0: 100% 1–5: 0%		49
rbc	*abnormal* 47 (44%)	*normal* 60 (56%)	*abnormal* 0 (0%)	*normal* 141 (100%)	152
pc	76 (39%)	118 (61%)	0 (0%)	141 (100%)	65
pcc	*present* 42 (17%)	*absent* 208 (83%)	*present* 0 (0%)	*absent* 146 (100%)	4
ba	22 (8.8%)	228 (91%)	0 (0%)	146 (100%)	4
htn	*yes* 147 (59%)	*no* 103 (41%)	*yes* 0 (0%)	*no* 150 (100%)	2
dm	137 (55%)	113 (45%)	0 (0%)	150 (100%)	2
cad	34 (14%)	216 (86%)	0 (0%)	150 (100%)	2
pe	76 (30%)	174 (70%)	0 (0%)	150 (100%)	1
ane	60 (24%)	190 (76%)	0 (0%)	150 (100%)	1
appet	*poor* 82 (33%)	*good* 168 (67%)	*poor* 0 (0%)	*good* 150 (100%)	1

Note: n (%). Miss. = number of missing values. Binary variables show both categories.
Abbreviations: sg = specific gravity; al = albumin (0–5 scale); su = sugar (0–5 scale); rbc = red blood cells; pc = pus cells; pcc = pus cell clumps; ba = bacteria; htn = hypertension; dm = diabetes mellitus; cad = coronary artery disease; appet = appetite; pe = pedal edema; ane = anemia.

4.1.2 CKD Disease Staging. The glomerular filtration rate (GFR) shows how well the kidneys are filtering waste and extra water from the blood [35]. Because measuring GFR directly requires time and is complex, healthcare professionals estimate it from a blood test by measuring creatinine levels. This is expressed as estimated glomerular filtration rate (eGFR). In this study, a modified eGFR* is used as an independent reference to understand whether the discovered patterns and groups follow meaningful differences in disease severity. The CKD dataset does not contain eGFR directly, and is missing a sex variable required by the standard CKD-EPI 2021 equation [34]. eGFR* is therefore computed from serum creatinine (sc) and age using a sex-agnostic adaptation of the CKD-EPI 2021 equation:

$$\text{eGFR}^* = 142 \cdot \min\left(\frac{\text{sc}}{\kappa}, 1\right)^{\alpha} \cdot \max\left(\frac{\text{sc}}{\kappa}, 1\right)^{-1.200} \cdot (0.9938)^{\text{age}} \quad (1)$$

The standard CKD-EPI 2021 equation uses sex-specific coefficients: $\kappa = 0.7, \alpha = -0.241$, and an additional multiplier of 1.012 for females, and $\kappa = 0.9, \alpha = -0.302$ for males. Since the dataset does not include a sex variable, parameters are set to midpoint values ($\kappa = 0.8$, $\alpha = -0.27$) as an unvalidated sex-agnostic approximation, and the female multiplier is omitted. The output is eGFR* in mL/min/1.73m^2. The original CKD-EPI equation is validated for adults aged 18 and older. A pediatric formula would be more appropriate for younger patients [43], but it requires height, which is not available in the dataset. For this reason, patients under 18 and those with missing age or serum creatinine are excluded from staging, which results in removing 19 pediatric and 25 unstageable patients, finally leaving 356 patients for staging. Since eGFR* is used only for post-hoc validation and not in the pipeline, these exclusions do not affect the core analysis.

The resulting eGFR* estimates for all patients are grouped into six categories following values for different stages of kidney function [35], as shown in Table 3.

Table 3: KDIGO categories for eGFR* staging.

Stage	Description	eGFR*
G1	Possible damage with **normal** kidney function	≥ 90
G2	Kidney damage with **mild loss** of function	60–89
G3a	**Mild to moderate** loss of kidney function	45–59
G3b	**Moderate to severe** loss of kidney function	30–44
G4	**Severe loss** of kidney function	15–29
G5	Kidney **failure**	< 15

Figure 1 shows the density of eGFR* values in CKD dataset by diagnosis. CKD patients are concentrated in the lower stages, with a peak around stage G5 and a long tail that extends through other stages. Non-CKD patients show a bimodal distribution with peaks around stages G2 and G1. The two distributions overlap through

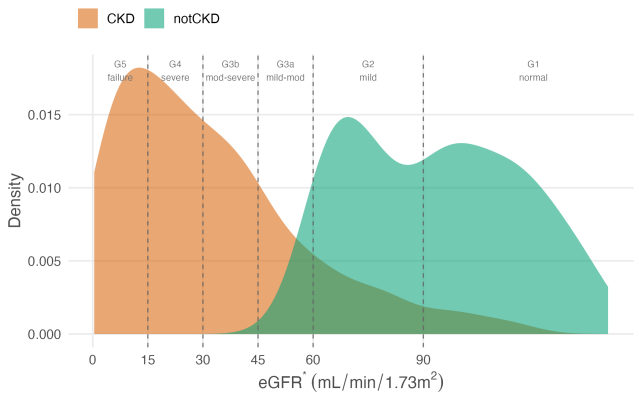


Figure 1: Distribution of eGFR* by diagnosis in CKD dataset.

normal and mild stages where eGFR* cannot reliably separate the two groups.

However, comparing missingness matrices for CKD and non-CKD patients, as shown in Figure 2 and Figure 3, exposes a clear difference for the stages shared by both groups. Blood count variables (rbc, rbcc, wbcc) and electrolytes (sodium, potassium) are missing far more often in CKD patients across all three stages. Additionally, at G3a, non-CKD patients have 0% missingness on every variable while CKD patients do not. So while eGFR* cannot separate the two groups in this range, missingness can.

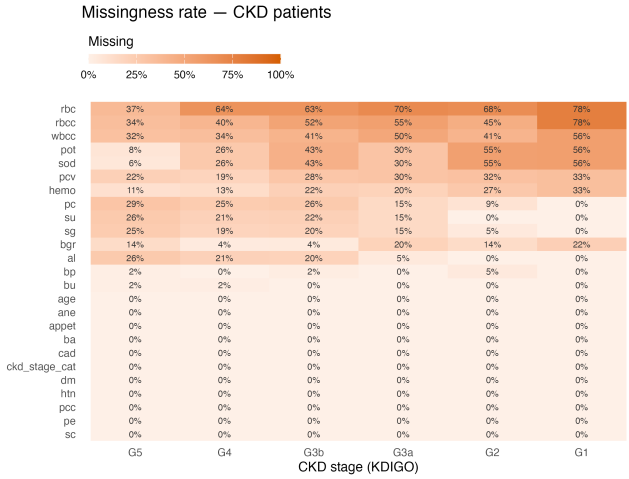


Figure 2: Missingness rate per clinical variable across KDIGO CKD stages for CKD patients. Variables are ordered by overall missingness rate.

4.1.3 Validation dataset. The Hepatitis dataset [3] from the UCI Machine Learning Repository is used as a second dataset to check whether the pipeline functions in a different setting. It has 155 patient records with 20 attributes: 6 continuous and 14 nominal, plus a binary survival outcome. Of the 155 records, 80 are complete. The highest missingness rates are in prothrombin time (43.2%) and alkaline phosphatase (18.7%).

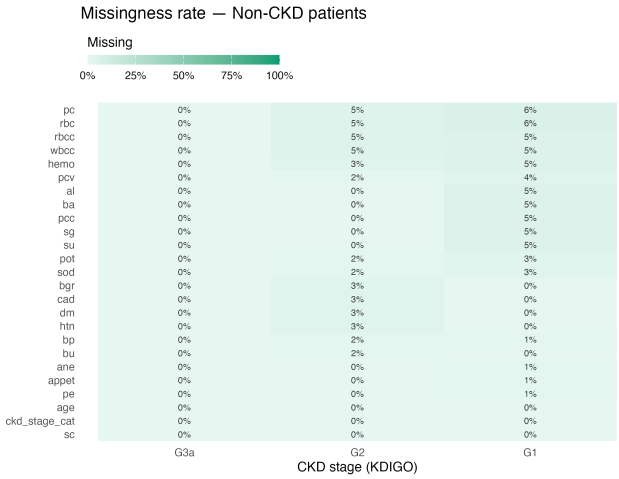


Figure 3: Missingness rate per clinical variable across KDIGO CKD stages for Non-CKD patients. Variables are ordered by overall missingness rate.

Table 4 and Table 5 show basic statistics for this dataset. Patients who died have higher bilirubin (median 1.95 vs 1.00 mg/dL) and lower albumin (3.30 vs 4.00 g/dL) and prothrombin time (39 vs 66 s) than survivors, consistent with worse liver function. Ascites, fatigue, and spider angiomata are also more common among patients who died (45% vs 5.0%, 94% vs 57%, and 71% vs 24%, respectively). The dataset is also imbalanced, with 123 survivors and 32 deaths, and skewed toward male patients (87% alive, 100% who died).

Table 4: Main statistics of Hepatitis dataset – continuous variables

Variable	Die (N = 32)	Live (N = 123)	Miss.
age	47 (39, 55) [30, 70]	38 (30, 50) [7, 78]	0
bilirubin	1.95 (1.20, 3.90) [0.40, 8.00]	1.00 (0.70, 1.20) [0.30, 4.60]	6
alk_phosphate	114 (85, 147) [62, 280]	85 (72, 126) [26, 295]	29
sgot	66 (48, 120) [16, 528]	55 (31, 92) [14, 648]	4
albumin	3.30 (2.60, 3.60) [2.10, 4.20]	4.00 (3.80, 4.30) [2.70, 6.40]	16
protime	39 (31, 46) [29, 90]	66 (52, 84) [0, 100]	67

Note: Median (Q1, Q3); [Min, Max] shown below. Miss. = number of missing values.
Abbreviations: age = age (years); bilirubin = total bilirubin (mg/dL); alk_phosphate = alkaline phosphatase (U/L); sgot = serum glutamic oxaloacetic transaminase (U/L); albumin = serum albumin (g/dL); protime = prothrombin time (s).

4.1.4 Hepatitis Dataset Staging. Albumin-Bilirubin (ALBI) grade is a simple model to assess liver function using only serum bilirubin and albumin levels [23]. The model was developed and tested on patients with chronic liver disease to eliminate the need for subjective variables needed in other grades. It has since been tested and

Table 5: Main statistics of Hepatitis dataset – categorical variables

Variable	Die (N = 32)		Live (N = 123)		Miss.
	male	*female*	*male*	*female*	
sex	32 (100%)	0 (0%)	107 (87%)	16 (13%)	0
steroid	20 (63%)	12 (38%)	56 (46%)	66 (54%)	1
antivirals	2 (6.3%)	30 (94%)	22 (18%)	101 (82%)	0
fatigue	30 (94%)	2 (6.3%)	70 (57%)	52 (43%)	1
malaise	23 (72%)	9 (28%)	38 (31%)	84 (69%)	1
anorexia	10 (31%)	22 (69%)	22 (18%)	100 (82%)	1
liver_big	3 (11%)	24 (89%)	22 (19%)	96 (81%)	10
liver_firm	13 (48%)	14 (52%)	47 (40%)	70 (60%)	11
spleen_palpable	12 (39%)	19 (61%)	18 (15%)	101 (85%)	5
spiders	22 (71%)	9 (29%)	29 (24%)	90 (76%)	5
ascites	14 (45%)	17 (55%)	6 (5.0%)	113 (95%)	5
varices	11 (35%)	20 (65%)	7 (5.9%)	112 (94%)	5
histology	7 (22%)	25 (78%)	78 (63%)	45 (37%)	0

Note: n (%). Miss. = number of missing values. Percentages computed over non-missing values within each group.
Abbreviations: liver_big = liver enlarged; liver_firm = liver firm on palpation; spleen_palpable = spleen palpable; spiders = spider angiomata; histology = abnormal histological finding.

used to evaluate hepatic conditions including chronic hepatitis B cirrhosis [10], chronic hepatitis C fibrosis [14], acute hepatitis B and D [8] etc. Here it is used as a post-hoc proxy while acknowledging the limitation that the grade boundaries originally derived may not carry the same meaning for this population. The ALBI score is computed as:

$$\text{ALBI} = 0.66 \cdot \log_{10}(\text{bilirubin}_{\mu\text{mol/L}}) - 0.0852 \cdot \text{albumin}_{\text{g/L}} \quad (2)$$

The UCI Hepatitis dataset reports bilirubin in mg/dL and albumin in g/dL. These are converted to the units required by the formula using the factors 1 mg/dL = 17.1 μmol/L and 1 g/dL = 10 g/L prior to scoring. Patients are assigned to one of three grades based on their ALBI score: Grade 1 (ALBI ≤ -2.60) for well-preserved liver function, Grade 2 ($-2.60 <$ ALBI ≤ -1.39) for moderate impairment and Grade 3 (ALBI > -1.39) for severe impairment [53]. The ALBI score has no formally published age restriction, but it was developed and validated on patients aged 18 and older[11]. This is the reason why during the staging patients younger than 18 were excluded and labeled as `Pediatric`. In this dataset, of the 155 patients, one patient aged 7 falls into this category. Patients that are missing bilirubin or albumin values are labeled as `Unscorable`, which applies to 16 patients, leaving 138 patients staged. The resulting ALBI grades are used to validate whether missingness patterns correspond to differences in liver state.

Figure 4 shows the density of ALBI scores by outcome. Patients who survived are concentrated in Grade 1 and Grade 2, with a peak just below -2.60. Patients who died are mostly concentrated in Grade 2 and Grade 3, with a peak in Grade 2 close to the Grade 3 boundary, and a tail extending into Grade 1. The two distributions overlap significantly in Grade 2, where ALBI grade cannot reliably separate the two outcomes.

Unlike the CKD dataset, the missingness matrices for the Hepatitis dataset (Figures 5 and 6) do not show a clear divide in the shared Grade 2 range. For example, protime and alkaline phosphatase are the variables missing most frequently in both cases,

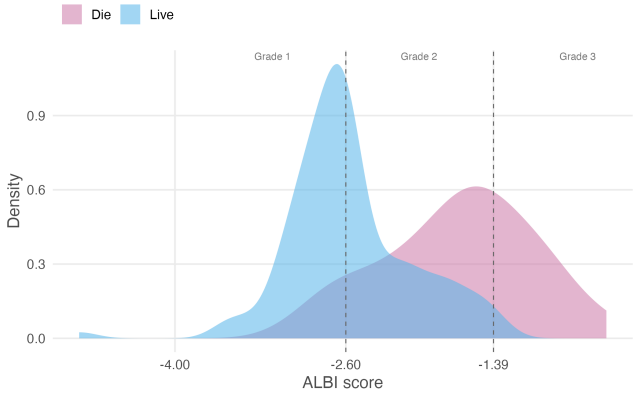


Figure 4: Distribution of ALBI score by outcome. Lower ALBI scores indicate better liver function.

and at similar rates (protime: 39% in Live vs. 31% in Die). Ascites, spiders, spleen and varices are more often missing in patients who died (6% vs. 2%), but the gap is significantly smaller than in the CKD dataset. This suggests that in comparison missingness doesn't have as much information for outcome in this dataset under the described assumptions.

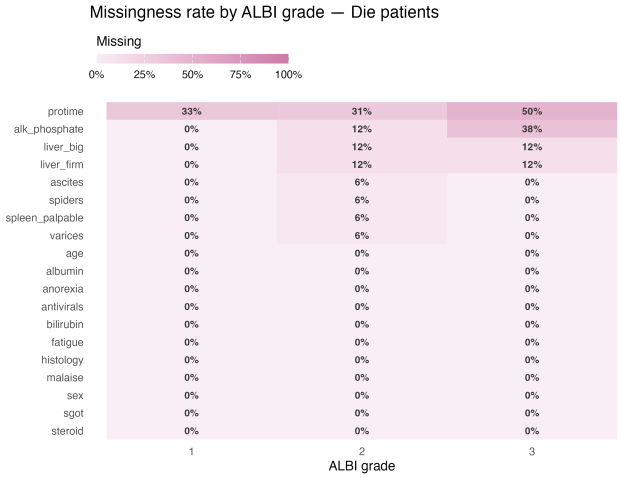


Figure 5: Missingness rate per clinical variable across ALBI grades for patients who died. Variables are ordered by overall missingness rate.

4.2 Missingness Metadata (RQ1)

In this subsection, we characterize missingness by identifying discrete patterns of variables that are missing together more often than expected if their missingness were independent. The goal is to have structured metadata elements that represent these patterns in a form that is readable. We also want to be able to model missingness for evaluation. First we represent the data using a missingness

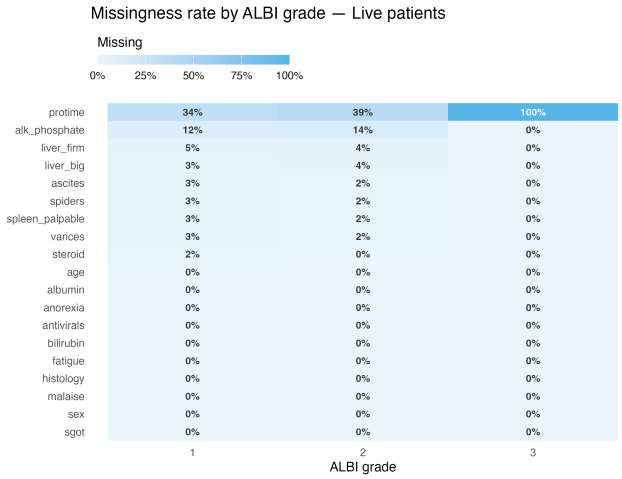


Figure 6: Missingness rate per clinical variable across ALBI grades for patients who survived. Variables are ordered by overall missingness rate.

indicator matrix. We then mine that matrix for combinations of variables that are missing together sufficiently often to be considered meaningful.

4.2.1 Missingness matrix. All analyses of missingness structure are done on a *missingness indicator matrix* [28]. It is a matrix of the form $M \in \{0,1\}^{n \times d}$, where n is the number of patients and d is the number of variables. More specifically, if x_{ij} is the value of variable j for patient i in the original data, M is defined as:

$$M_{ij} = \begin{cases} 1, & \text{if } x_{ij} \text{ is missing} \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Only columns with at least one missing value are kept, leaving $d' \leq d$ variables and a reduced matrix $M' \in \{0,1\}^{n \times d'}$.

4.2.2 Co-missingness mining. Every patient is represented by a set of variables that are missing in their record, i.e. the subset of columns in M' for which the entries in that row are equal to 1. Each such set is called a *transaction*. Any subset of the d' variables is called an *itemset* [4]. An itemset S occurs in a transaction if all variables in S are missing for that patient. The frequency at which an itemset appears in transactions is called *support*. More precisely, for an itemset S over the columns of M', support is defined as:

$$\text{support}(S) = \frac{1}{n} \sum_{i=1}^{n} \prod_{j \in S} M'_{ij}. \tag{4}$$

The product is 1 only when every column in S is missing in row i, and 0 otherwise. An itemset is *frequent* if its support meets a predefined minimum support threshold. Only frequent itemsets are kept, regardless of size.

Frequent itemsets are mined using the *Apriori algorithm* [5], as implemented in [36]. Apriori relies on a simple property, known as the *downward closure property*, which states that if an itemset is frequent, every subset of it must be frequent too. Apriori returns frequent itemsets grouped by size, starting with all single columns, then all pairs, and so on. Due to the downward closure property, reporting every frequent itemset would repeat the same group of variables multiple times. To reduce this redundancy, only *maximal itemsets* are kept. Those are the ones that are not a subset of a larger frequent itemset. Single-column itemsets are not included, and only those of size two or more are considered.

4.2.3 Missingness block detection. For a maximal itemset S, we use *any-confidence* [37], defined as:

$$\text{any-confidence}(S) = \frac{\text{support}(S)}{\min_{c \in S} \text{support}(\{c\})}. \tag{5}$$

Any-confidence gives a ratio of the support of an itemset to the support of its least frequently missing member. A value close to 1 means that whenever the least frequently missing member is missing, the full itemset is almost always missing as well.

Any-confidence is closely related to *all-confidence* measure [37], defined using the most frequently missing member instead. More precisely, we define:

$$\text{all-confidence}(S) = \frac{\text{support}(S)}{\max_{c \in S} \text{support}(\{c\})}. \tag{6}$$

All-confidence gives a formal lower bound on the confidence of every single-member rule that can be derived from S. When a member of S is frequently missing outside the group, all-confidence is low, even if the other members are almost always missing together.

Unlike all-confidence, any-confidence does not satisfy the downward closure property. For this reason, it is not suitable as a constraint for itemset mining [37]. This limitation does not affect the mining procedure used here, since Apriori first identifies frequent itemsets using support. Any-confidence is then applied to the maximal frequent itemsets as a post-mining filtering that helps identify relevant blocks.

Since each measure focuses on only one extreme of the itemset, we use the *cross-support ratio* [51] to describe the difference between these reference members. The measure is defined as:

$$\text{cross-support}(S) = \frac{\text{all-confidence}(S)}{\text{any-confidence}(S)} = \frac{\min_{c \in S} \text{support}(\{c\})}{\max_{c \in S} \text{support}(\{c\})}. \tag{7}$$

A value close to 1 means that the least and the most frequently missing members of the itemset have similar individual missingness rates. Therefore, the members of the itemset have similar missingness frequencies overall. A lower value of cross-support requires a closer inspection of the pattern.

Itemsets for which any-confidence meets a chosen minimum threshold, are reported as *relevant blocks*. Defining a relevant block, and therefore choosing a mimimum threshold, is a decision made by a researcher and depends on the dataset. A higher threshold results in fewer reported blocks, while a lower threshold results in more. Together with Apriori's minimum support threshold, the any-confidence threshold can be modified to explore the structure of missingness.

For each relevant block S, we also compute the probability that the entire block is missing, given that a particular member is missing:

$$P(S \mid c \text{ is missing}) = \frac{1}{|\{i : M'_{ic} = 1\}|} \sum_{i : M'_{ic} = 1} \prod_{j \in S} M'_{ij}. \tag{8}$$

This probability describes how strongly each member is associated with the rest of the block. If the values are similar for all members, the block has a consistent structure.

The resulting descriptions of the identified relevant blocks are reported together with their mining metrics. For each block, the pipeline reports its any-confidence, all-confidence, cross-support ratio, and the conditional probability of the complete block missing given that each member is missing. These values describe both the strength of the co-missingness pattern and the contribution of individual variables to the block. Variables that are not assigned to a relevant block are reported individually using their missingness support.

4.2.4 Missingness patterns. The relevant blocks identified in Subsection 4.2.3 are used to define missingness patterns. Each pattern is named based on the combination of variables that are missing together. A new column is added to the dataset containing the missingness pattern assigned to each patient. Patients who match multiple patterns receive a combined label containing the corresponding variable combinations. Fully observed patients are labelled `mp0`, while patients with missing values that do not match any of the identified patterns are labelled `mp_other`.

4.2.5 Missingness predictability. For each column with missing values, we train a random forest classifier to predict whether a value is missing using the observed values of all remaining variables. We use this to determine whether the missingness of a variable can be predicted from the other observed variables. If it can, this provides evidence against the MCAR assumption. However, this does not distinguish between MAR and MNAR.

We do not include missingness indicators as predictors because the model could then rediscover the missingness patterns directly. Missing values in the predictor variables are replaced by random draws from the corresponding observed values. This avoids introducing information about missingness through fixed imputed values such as the median. Categorical variables are one-hot encoded after imputation.

Columns with fewer than 5 missing or non-missing observations are not included. We chose this minimum based on the primary dataset (Subsection 4.1.1) and the number of folds used for validation. With fewer observations, some validation folds could contain no observations from one of the two classes. We use the same threshold for the Hepatitis dataset without further adjustment (Subsection 4.1.3).

We repeat the experiment 20 times to reduce the effect of the random imputation. This provides multiple random imputations while keeping the computational cost manageable. We evaluate the predictor using AUC for each fold. We use AUC because the target is binary and the proportion of missing values can differ substantially between variables. AUC measures how well the model distinguishes between missing and non-missing observations based on the predicted missingness probabilities.

For each column, we report the mean AUC and standard deviation across folds and repetitions. A higher AUC indicates that the missingness of the variable is more predictable from the observed values of the other variables. We also report the three predictors with the highest mean feature importance for each column. These predictors show which observed variables contribute most to predicting whether the target variable is missing.

4.3 Synthetic Data Generation With Missingness (RQ2)

The goal of RQ2 is to examine whether representing the missingness structure of the original data affects the similarity of the synthetic data to the real data without reducing downstream predictive performance. We compare two missingness representation strategies within the same synthesis pipeline. Both strategies use the same synthesis model, the same pre-synthesis imputation procedure, and the same downstream evaluation procedure.

4.3.1 Synthesizer. We generate the synthetic data using a Bayesian network model [6, 7]. The model supports both discrete and continuous variables and provides two options for modelling continuous variables: `LinearGaussianCPD` and `CKDE`. In our configuration, we enable only the arc-search operators. These operators determine the network structure, while the node-type operator, which would allow the model to choose between different continuous-node models, is disabled. As a result, all continuous nodes are modelled using `LinearGaussianCPD`.

Each discrete node is fitted as a `DiscreteFactor` and can only have discrete parents. We learn the network structure using greedy hill-climbing with the Bayesian Information Criterion (BIC) as the scoring function. The search can add, remove, or reverse arcs, with a maximum of two parents per node. We do not blacklist or whitelist any arcs, so the search can learn dependencies between the variables in the training data. The configuration used in the experiments is summarized in Table 6.

Table 6: Synthesizer configuration for the experiment.

Component	Setting
Network type	Semiparametric (`SemiparametricBNType`)
Continuous node factors	`LinearGaussianCPD`
Discrete node factors	`DiscreteFactor`
Structure search	Greedy hill-climbing (`hc`)
Scoring function	BIC
Search operators	Arc addition, removal, reversal
Maximum in-degree	2
Arc blacklist / whitelist	None
Random seed	42 by default
Sample size	Equal to the training set size

Structure learning is not fully deterministic given a fixed random seed. Repeated hill-climbing runs on identical data can converge to different structures. This is a property of the search procedure itself and it affects the interpretation of the results of the experiment (Section 5.2) and the subgroup analysis (Section 5.3).

4.3.2 Imputation. Before synthesis, missing values are imputed in the training data. We use column-wise imputation as a design choice to provide the same complete input format to both missingness representation strategies. The current pipeline uses column-wise imputation to fill missing values. For each variable j, the imputed value $\tilde{x}_{ij}$ is defined as:

$$\tilde{x}_{ij} = \begin{cases} x_{ij}, & \text{if } M'_{ij} = 0, \\ \text{median}\{x_{ij} \mid M'_{ij} = 0\}, & \text{if } M'_{ij} = 1 \text{ and } j \text{ is continuous}, \\ \text{mode}\{x_{ij} \mid M'_{ij} = 0\}, & \text{if } M'_{ij} = 1 \text{ and } j \text{ is categorical}. \end{cases} \tag{9}$$

The median and mode are computed using the observed values in the corresponding training fold (Subsection 4.3.3). After synthesis, the missingness pattern is reconstructed, so the imputed values are not included in the final synthetic dataset.

4.3.3 Pipeline Design. Figure 7 shows one fold of the evaluation pipeline used in this experiment. Let $R = (X, y)$ be the real dataset. For each cross-validation fold k, R is split into a training set $R^{(k)}_{\text{train}}$ and a test set $R^{(k)}_{\text{test}}$. Only $R^{(k)}_{\text{train}}$ is used for synthetic data generation, while $R^{(k)}_{\text{test}}$ remains unchanged and is used only for evaluation. This is TSTR methodology introduced in the Section **??**. Generating synthetic data from incomplete training data in our set up requires handling missingness before and after synthesis.

Before synthesis, $R^{(k)}_{\text{train}}$ is imputed, producing a completed training set $\tilde{R}^{(k)}_{\text{train}}$. The synthesis model trains on this fully observed data and generates a fully observed synthetic training set $S^{(k)}_{\text{raw}}$. Because $S^{(k)}_{\text{raw}}$ does not contain missing values, missingness is reconstructed afterward, producing the final synthetic training set $S^{(k)}_{\text{train}}$. A downstream prediction model is then trained on $S^{(k)}_{\text{train}}$ and evaluated on the corresponding real test set $R^{(k)}_{\text{test}}$.

The design decision within the synthesis pipeline concerns how missingness is represented before and reconstructed after synthesis. We compare two approaches: *shadow columns* (Subection 4.3.4), which represent missingness independently for each variable, and *shadow patterns* (Subsection 4.3.5), which represent the joint missingness structure identified using methods in Section 4.2. Both approaches use the same simple imputation procedure before synthesis (Subsection 4.3.2) and the same synthesis model (Subsection 4.3.1). The differences in the two synthesis strategies are summarized in Table 7.

Table 7: Overview of the two RQ2 strategies for representing and reconstructing missingness during synthesis. Both strategies use the same imputation and synthesis model.

Condition	Missingness representation
Shadow columns (*S1*)	One binary indicator per variable; missingness is reconstructed independently for each variable (Subsection 4.3.4).
Shadow patterns (*S2*)	One categorical pattern label representing co-missingness blocks; missingness is reconstructed from the sampled pattern and residual rates (Subsection 4.3.5).

Condition *R* serves as the real-data baseline. No synthesis is performed, so the downstream prediction model is trained directly on the real training data $R^{(k)}_{\text{train}}$ and evaluated on $R^{(k)}_{\text{test}}$. Conditions *S1* and *S2* instead train the same downstream prediction model on the corresponding synthetic training set $S^{(k)}_{\text{train}}$ and evaluate it on the same real test set $R^{(k)}_{\text{test}}$. In all conditions, the downstream training procedure uses MICE followed by logistic regression. Because every condition uses the same evaluation procedure, observed differences in downstream performance can be attributed to the missingness-handling strategy used to produce $S^{(k)}_{\text{train}}$. The experiment is repeated across the k cross-validation folds, which provides multiple training and test splits and reduces the influence of instability in the synthesis. The test set $R^{(k)}_{\text{test}}$ is never used during synthesis or synthetic data generation.

Cross-validation is stratified on y so that class balance is preserved across folds, with a fixed random seed for reproducibility. Block mining, simple imputation, synthesis, and missingness reconstruction are all performed using only $R^{(k)}_{\text{train}}$ within each fold. In Section 5.1, we report the missingness patterns mined from the full dataset to provide an overall description of the observed missingness structure. These full-dataset patterns are not used during the RQ2 evaluation. Instead, the block-mining procedure is repeated within each training fold, using only $R^{(k)}_{\text{train}}$, to avoid information leakage. The fold-specific mined patterns are not reported individually in the context of the thesis, but are available after the synthesis as exported structural metadata element.

4.3.4 Missingness: shadow columns. Condition *S1* represents missingness independently by assigning one binary shadow indicator to each variable. Before synthesis, the missingness indicator matrix $M^{(k)}_{\text{train}}$ is appended column-wise to the imputed training data $\tilde{R}^{(k)}_{\text{train}}$. This adds one binary shadow column for each variable. The synthesis model then learns the joint distribution of the clinical variables, the outcome, and the missingness indicators.

After synthesis, the sampled shadow indicators M^{raw}_{ij} are used to reconstruct missing values. For each synthetic observation i and variable j,

$$x_{ij} \leftarrow \text{NaN} \qquad \text{if } M^{\text{raw}}_{ij} = 1. \tag{10}$$

The shadow columns are then removed, leaving the final synthetic dataset with the original clinical variables and outcome.

4.3.5 Missingness: shadow patterns. Condition *S2* represents missingness using the co-missingness patterns identified using the methods described in Section 4.2. The pattern-labeling scheme and labels such as `mp0` and `mp_other` are defined in Subsection 4.2.4. As mentioned in the Subsection 4.3.3, within each training fold, the missingness blocks are mined using only the training partition. Each training sample is then assigned a pattern label according to these fold-specific blocks. The pattern label is added to the imputed training data as an additional categorical variable and is learned jointly with the clinical variables and outcome during synthesis.

After synthesis, the generated pattern is used to reconstruct the corresponding co-missingness block. Missingness outside these blocks is then added using residual missingness rates estimated from the corresponding real training fold.

Residual rates. The mined blocks represent the structured component of the missingness mechanism. Missingness that is not explained by these blocks is added using residual missingness rates estimated from the real training data. These rates are calculated

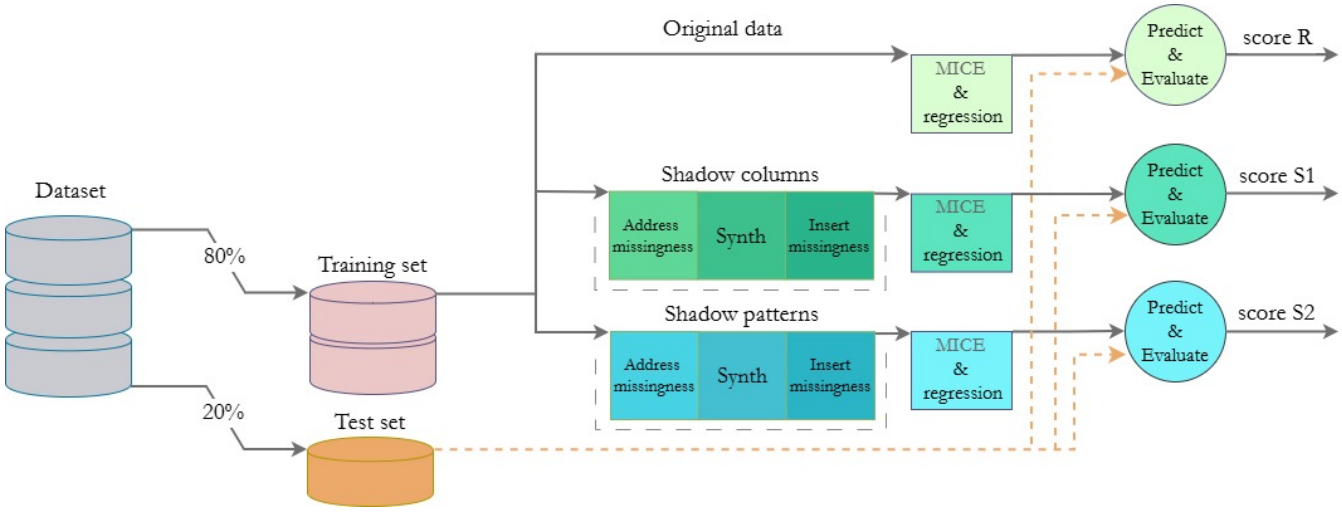


Figure 7: ***Pipeline for a single fold: training data are imputed and used to fit both synthesis branches (S1 and S2). The resulting real and synthetic datasets are then evaluated for downstream utility and privacy.***

separately for each variable and outcome class. For variables that belong to a mined block, the residual rate is estimated only from observations in which the corresponding block is not present, i.e. where the variables in that block are not all missing. For variable j and outcome class c, this rate is defined as:

$$r_{j,c} = P(x_j \text{ missing} \mid \text{corresponding block not present}, y = c). \tag{11}$$

For variables that do not belong to a mined block, the residual rate is their overall class-specific missingness probability:

$$r_{j,c} = P(x_j \text{ missing} \mid y = c). \tag{12}$$

These rates represent missingness that occurs outside the identified co-missingness blocks. They are estimated from the training data and applied after the deterministic missingness patterns have been reconstructed.

Missingness in the synthetic data is reconstructed in two stages. First, the generated missingness pattern determines which variables are set to missing. For example, if an observation is assigned a pattern containing variables j and k, both variables are set to missing. Second, residual missingness is added to observations whose generated pattern is not `mp0`. For each variable j, the probability of adding residual missingness depends on the generated outcome class y_i and the corresponding class-specific residual rate $r_{j,c}$. For variables that belong to a mined co-missingness block, residual missingness is only considered if the block was not already triggered by the generated pattern. This prevents the same variable from being made missing twice. Variables outside the mined blocks can receive residual missingness whenever the generated pattern is not `mp0`. The pattern-based and residual missingness are then combined to produce the final synthetic dataset used for downstream evaluation.

4.3.6 Downstream evaluation. The downstream evaluation measures predictive performance using several metrics. AUC measures how well the model distinguishes between the two outcome classes, while the Brier score measures how close the predicted probabilities are to the observed outcomes. However, AUC does not show how well the model performs when a specific classification threshold is used. Two models with similar AUC can therefore have different sensitivity, specificity, precision, and F1 scores.

For these classification-based metrics, we use a fixed threshold of 0.5. A patient is classified as positive if the predicted probability is at least 0.5 and as negative otherwise. We use the same threshold for all three conditions (*R*, *S1*, and *S2*), across all folds and for both datasets. We do not tune the threshold separately for each condition.

4.3.7 Pattern preservation in generated datasets. Pattern preservation is evaluated by comparing the distribution of missingness patterns between the real and synthetic datasets. Following Wang, Asif, and Vaidya [49], the divergence in missingness-pattern distribution is defined as the L_1 distance between the two pattern distributions:

$$D_{\text{mis}}(A, A') = \|P_A - P_{A'}\|_1 = \sum_{m \in \mathcal{MP}} |P_A(m) - P_{A'}(m)|, \tag{13}$$

where $\mathcal{MP}$ is the set of missingness patterns, and $P_A(m)$ and $P_{A'}(m)$ are the proportions of observations assigned to pattern m in datasets A and A', respectively. A lower D_{mis} indicates greater similarity between the missingness-pattern distributions.

This evaluation is performed independently of the downstream evaluation (Subsection 4.3.6). Each synthesis branch (*S1*, *S2*) is run 16 times using different random seeds. Each run produces one synthetic dataset with the same number of rows as the real dataset. D_{mis} is computed for each synthetic draw against the pattern distribution of the real data. Repeated runs are used because PyBNesian's hill-climbing structure search is not fully deterministic, as described in Subsection 4.3.1.

4.4 Subgroup Analysis RQ3

In this subsection, we compare the real and synthetic data for different subgroups. We consider subgroup-specific predictive performance and privacy exposure. The subgroup definitions are defined using the real data and are then applied consistently during the evaluation. The results are compared between the real data and the two synthetic conditions.

4.4.1 Subgroup definitions. Subgroups are defined using the real data and kept fixed throughout the evaluation. We consider two types of subgroups: clustering-based outliers and predefined subgroups. For the clustering-based subgroup, patients are first partitioned using k-means clustering. Before clustering, missing values are imputed as described in Subsection 4.3.2, categorical variables are encoded, and all features are standardized. We use $k = 2$ clusters, which was chosen through a manual comparison of candidate values of k outside the evaluation pipeline. The clustering is repeated with ten different random initializations, and the mean ARI is reported as an indicator of stability. Silhouette width is also reported as a diagnostic of cluster separation.

The resulting clusters are described using descriptive statistics. A shallow decision tree is fitted to provide a readable description of the features that distinguish the clusters. The tree is used for description only and does not determine cluster membership. Outliers are then identified within the resulting clusters using the distance to the assigned cluster centroid. For each cluster, the mean and standard deviation of these distances are calculated separately. A patient is classified as an outlier when their distance is more than two standard deviations above the cluster-specific mean. The resulting outlier rule is defined on the real data before cross-validation and kept fixed throughout the evaluation. The same rule is applied to synthetic patients. A shallow decision tree is also fitted to provide a readable description of the resulting outlier rule.

The second type of subgroup consists of predefined subgroups specified in the dataset configuration. These include groups based on characteristics such as age and comorbidity status, as well as a data-completeness group defined by the absence of missing values. These subgroup definitions are also fixed using the real data and applied unchanged to the synthetic data.

4.4.2 Subgroup-specific predictive evaluation. Predictive performance is evaluated using the same cross-validation procedure described in Section 4.3.6. Predictions for each subgroup are collected as out-of-fold (OOF) predictions across all folds. This gives each patient one prediction for each evaluation condition. Predictions are pooled across folds before subgroup performance is calculated. This provides a single subgroup estimate based on all available out-of-fold predictions.

For each subgroup, the True Positive Rate (TPR) and True Negative Rate (TNR) are calculated separately for the subgroup and the remaining patients. For a group g, these are defined as:

$$\mathrm{TPR}_g = \frac{TP_g}{TP_g + FN_g}, \qquad \mathrm{TNR}_g = \frac{TN_g}{TN_g + FP_g}, \qquad g \in \{\mathrm{sub}, \mathrm{rest}\}. \tag{14}$$

The corresponding subgroup disparities are defined as

$$\Delta_{\mathrm{TPR}} = \mathrm{TPR}_{\mathrm{rest}} - \mathrm{TPR}_{\mathrm{sub}}, \qquad \Delta_{\mathrm{TNR}} = \mathrm{TNR}_{\mathrm{rest}} - \mathrm{TNR}_{\mathrm{sub}}. \tag{15}$$

A positive Δ_{TPR} therefore indicates lower sensitivity for the subgroup than for the remaining patients. Similarly, a positive Δ_{TNR} indicates lower specificity for the subgroup. Together, these metrics describe differences in the two components of equalized odds [20].

The real-data disparities provide the reference for comparison with synthetic data. For each synthetic condition, the change relative to the real data disparity is calculated as

$$D_{S1} = \Delta_{S1} - \Delta_R, \qquad D_{S2} = \Delta_{S2} - \Delta_R. \tag{16}$$

The original Δ values are reported together with these differences, since a change in sign may indicate that the direction of the disparity has reversed.

The TPR gap is defined when both the subgroup and the remaining population contain at least one actual positive case. The TNR gap similarly requires at least one actual negative case in both groups. When one of these outcome classes is completely absent, the corresponding metric is undefined and is reported as such. A small but nonzero number of cases does not make the metric undefined, although the resulting estimate is less reliable.

Subgroup disparity estimates are computed independently for each of several repeated synthesis runs. We use 3 independent runs for CKD and 5 for Hepatitis dataset. The resulting estimates are summarized across runs using the mean and standard deviation.

4.4.3 Reliability of subgroup results. Subgroup estimates are reported together with the subgroup size and, where relevant, the number of positive and negative cases. This is important because the subgroup metrics are empirical proportions and can be sensitive to small numbers of outcome cases. Differences between conditions are therefore interpreted as differences in the observed point estimates.

Predictive performance, privacy exposure, and representation are computed from independently generated synthetic datasets, using separate synthesis batches. This means that the metrics are treated as separate estimates of properties.

4.4.4 Subgroup-specific privacy evaluation. Privacy exposure is evaluated using a singling-out attack based on the Anonymeter framework [17]. A singling-out attack attempts to identify individual records using combinations of attributes. The evaluator generates attack queries from the synthetic data and evaluates their success against the real training data and a held-out control population.

For each fold and synthetic condition, the real training data is used as the original data, the synthetic data as the attack resource, and the held-out real data as the control population. The aggregate Anonymeter risk is recorded for each fold. The difference between attack success on the training and control populations provides an estimate of the extent to which successful attacks may be associated with the training data and not occurring by chance [17]. To examine whether privacy exposure differs between subgroups, the attack queries that successfully identify records in the real training data are applied to the held-out control population. Each control-set patient identified by at least one of these queries is recorded. The

previously defined subgroup rules are then applied to the identified patients without refitting. The subgroup attack rate is calculated as the proportion of patients in each subgroup that are identified by the attack and is compared with the corresponding rate for the remaining patients.

Because Anonymeter's attack-query generation is not seeded, the subgroup attack rates can vary between repeated runs of the same experiment. For each dataset we use five independent runs and report privacy results using the mean and standard deviation. The subgroup level rates are interpreted as estimates of relative exposure.

5 EVALUATION

This chapter evaluates the two missingness representation strategies described in Chapter 4. The evaluation follows the three research questions. First, we examine the missingness structure identified in the two datasets (RQ1). Second, we compare the two strategies in terms of downstream predictive performance and missingness-pattern preservation (RQ2). Finally, we examine subgroup-specific predictive performance and privacy exposure (RQ3).

5.1 Missingness Metadata (RQ1) Results

In this subsection, we show and describe the relevant blocks identified using the missingness design. In the evaluation, we consider two criteria. First, if the block corresponds to a recognizable clinical test panel. Second, if the missingness of a block is associated with defined stages or disease categories. Since minimum support for Apriori and minimum any-confidence for filtering are chosen by a researcher, we describe the choice of specific parameters. We do the same for both datasets.

5.1.1 Missingness patterns in CKD dataset. Before choosing a minimum support in Apriori, we check the distributions of individual missingness rates, as shown in Table 9. The rates show a clear drop between variables missing in at least 11% of patients (e.g. `bgr` at 0.11) and those missing in under 5% (e.g. `bu` at 0.048). A threshold of 0.1 therefore separates variables in what we decide is substantial missingness from those that are rarely missing. After running initial tests with varying values from $[0, 1]$ and using different increments, we confirmed this threshold as our primary choice. One of the reasons is that below this value, we extract blocks defined by extremely small subgroups of patients. For example, at the minimum support of 0.01, a `[ba,pcc]` block is identified with any-confidence equal to 1.0, but it belongs to 4 non-CKD patients that are absent from all KDIGO categories or CKD severity groups. Another reason is that the number of reported patterns rises sharply below this value. At minimum support of 0.05, 0.04, 0.03, 0.02, and 0.01, the pipeline reports 8, 12, 15, 31, respectively, before fully exploding for 0.01.

Next, we vary the choice of minimum any-confidence threshold, which filters maximal frequent itemsets. At 0.7, three blocks clear the threshold: `[pot,sod]`, `[al,pc,rbc,sg,su]`, and `[hemo,pcv,rbcc,wbcc]`, with any-confidence of 1.000, 0.891, and 0.865 respectively. Lowering the threshold to 0.6 additionally extracts two patterns `[pot,rbc,sod]` and `[pcv,rbc,rbcc,wbcc]`, with any-confidence values of 0.621 and 0.606, respectively. These additional blocks share the variables with the higher-confidence blocks. They are a more detailed representation of the missingness structure, but they also introduce more overlap between patterns. This can make the interpretation more difficult. At 0.4, a sixth block, `[pot,rbcc,sod,wbcc]`, is reported, which adds to the problem of overlapping patterns.

At the minimum support of 0.1 and minimum any-confidence 0.7, we report the three relevant blocks mentioned earlier. One represents an electrolyte panel (potassium, sodium), another a urinalysis panel (albumin, pus cells, red blood cells, specific gravity, sugar), and the third a complete blood count panel (haemoglobin, packed cell volume, red blood cell count, white blood cell count). The three blocks share no variables.

All-confidence and cross-support values, however, differ from any-confidence for two of the three blocks. This shows that the urinalysis and complete blood count blocks are not internally uniform, while the electrolyte panel is. The electrolyte panel has an all-confidence and cross-support ratio of 0.989, close to its any-confidence of 1.000, indicating both members are missing together almost every time either one is. The urinalysis block has an all-confidence of 0.270 and a cross-support ratio of 0.303. The complete blood count block has an all-confidence of 0.344 and a cross-support ratio of 0.397. In both of these cases, one member is missing far less reliably. The per-column conditionals show this directly. In the urinalysis block, `al`, `sg`, and `su` have conditionals above 0.83, while `rbc` has a conditional of only 0.270. In the complete blood count block, `hemo` has a conditional of 0.865, while `rbcc` has a conditional of only 0.344. This indicates that `rbc` and `rbcc` are missing together with the rest of their respective blocks far less consistently than other members. Restricting the urinalysis block to its three most reliable members, `al`, `sg`, and `su`, gives any-confidence of 0.957, and all-confidence of 0.898. This indicates that there is a tighter core within the block. We report this as a point for interpretation, but we do not analyze these internal structures further in this work.

The three blocks also overlap little with one another when we look at the patients behind them. Of the 87 patients missing the electrolyte panel, 9 (10.3%) also miss the urinalysis block (22.0% of the 41 patients missing urinalysis), and 19 (21.8%) also miss the complete blood count block (42.2% of the 45 patients missing CBC). Only 5 patients miss all three blocks at once. For the urinalysis and complete blood count blocks specifically, of the patients missing at least one, only 22–24% are also missing the other (10 of 41 patients missing the urinalysis block, and 10 of 45 patients missing the complete blood count block). This suggests that the three blocks represent mostly separate missingness patterns. All ten patients missing both the urinalysis and complete blood count blocks are CKD patients, and 82 of the 87 patients missing the electrolyte panel (94.3%) are CKD patients as well. The electrolyte panel is missing in 32.8% of CKD patients and only 3.3% of non-CKD patients. The urinalysis block is missing in 14.8% of CKD patients and 2.7% of non-CKD patients. The complete blood count block is missing in 16.8% of CKD compared to 2.0% for non-CKD.

If we look closely at CKD patients, as detailed in Table11, the three blocks show different patterns in relation to disease severity. The urinalysis block is missing in 21.5%, 18.9%, and 19.6% of patients at KDIGO stages G5, G4, and G3b respectively, decreasing to 5% at G3a and 0% at G2 and G1. In contrast, the complete blood count block shows the opposite pattern, with higher missingness in less severe CKD stages, increasing from 10.8% at G5 to 27.3% at G2 and 33.3% at G1. The electrolyte panel follows neither pattern cleanly:

it is lowest at G5 (6.2%), rises sharply through G4 (26.4%) to a local peak at G3b (43.5%), dips slightly at G3a (30.0%), then rises again to its highest values at G2 (54.5%) and G1 (55.6%). Broadly, then, it behaves more like the complete blood count block than the urinalysis block, meaning missingness is lowest among the most severely affected patients and highest among the mildest, but without a clean monotonic trend across the intermediate stages.

The same broad pattern can also be seen using the broader disease categories as show in Table 8. Urinalysis missingness decreases from 21.5% in Failure to 16.8% in Disease and 0% in both Early and Normal stages. Complete blood count missingness increases from 10.8% in Failure to 15.1% in Disease, 27.3% in Early, and 33.3% in Normal. Electrolyte panel missingness increases even more sharply across the same categories, from 6.2% in Failure to 33.6% in Disease, 54.5% in Early, and 55.6% in Normal. These broader categories provide a simplified view of the same distribution across disease severity. They help interpret the observed patterns: urinalysis missingness is more common for patients with advanced CKD, while complete blood count and electrolyte-panel missingness are both more common for patients with less severe or no apparent CKD, though the electrolyte panel shows this effect more strongly than either of the other two blocks.

Table 8: Missingness rates for relevant blocks across broader CKD disease categories.

Block	Failure	Disease	Early	Normal
Urinalysis	21.5%	16.8%	0.0%	0.0%
Complete blood count	10.8%	15.1%	27.3%	33.3%
Electrolyte panel	6.2%	33.6%	54.5%	55.6%

5.1.2 Missingness predictability in CKD dataset. The missingness predictability results for CKD are shown in Table 10. The variables in the identified missingness patterns (Subsection 5.1.1) generally have high AUC values. For missingness patter, containing `pot` and `sod`, the AUC values are 0.787 and 0.792, respectively. The variables in the second pattern have AUC values varying between 0.749 for `pc` and 0.865 for `rbc`. For the third pattern, the AUC values vary between 0.756 for `hemo` and 0.847 for `rbcc`. All remaining variables have mean AUC values above 0.5, with the lowest value observed for `age` at 0.530. The standard deviation for `age` is 0.069, indicating more variation between repetitions than for most other variables. The highest AUC values are observed for `rbc` (0.865), `rbcc` (0.847) and `wbcc` (0.827).

The feature importance results show that `hemo`, `sc`, `pcv` and `rbcc` occur frequently among the three highest-ranked predictors. `hemo` is among the top three predictors for 12 of the 16 variables, while `sc` is among the top three for 9 variables. Several variables that do not belong to the identified missingness patterns also have relatively high AUC values, including `bp` (0.741), `bgr` (0.731) and `sc` (0.659). The standard deviations range from 0.011 to 0.071.

Table 9: Missingness rates for all variables in the CKD and Hepatitis datasets. Dashed line represents the chosen minimum support threshold for Apriori.

CKD		Hepatitis	
Variable	Missing (%)	Variable	Missing (%)
rbc	38.0	protime	43.2
rbcc	32.8	alk_phosphate	18.7
wbcc	26.5	albumin	10.3
pot	22.0	liver_firm	7.1
sod	21.8	liver_big	6.5
pcv	17.8	bilirubin	3.9
pc	16.3	spiders	3.2
hemo	13.0	ascites	3.2
su	12.3	varices	3.2
sg	11.8	spleen_palpable	3.2
al	11.5	sgot	2.6
bgr	11.0	anorexia	0.6
bu	4.8	fatigue	0.6
sc	4.3	steroid	0.6
bp	3.0	malaise	0.6
age	2.3	sex	0.0
pcc	1.0	age	0.0
ba	1.0	antivirals	0.0
htn	0.5	histology	0.0
dm	0.5		
cad	0.5		
appet	0.3		
pe	0.3		
ane	0.3		

Table 10: Predictability of missingness in the CKD dataset.

Variable	Missing	Mean AUC	SD	Top 3 predictors
rbc	152	0.865	0.011	hemo, sc, pcv
rbcc	131	0.847	0.014	hemo, pcv, sc
wbcc	106	0.827	0.020	rbcc, hemo, pcv
su	49	0.821	0.021	sc, hemo, rbcc
al	46	0.803	0.023	sc, rbcc, hemo
sg	47	0.793	0.026	sc, hemo, rbcc
sod	87	0.792	0.015	hemo, sc, bu
pot	88	0.787	0.015	hemo, sc, sod
pcv	71	0.773	0.024	hemo, rbcc, bu
hemo	52	0.756	0.029	pcv, rbcc, bu
pc	65	0.749	0.024	bgr, sc, hemo
bp	12	0.741	0.048	age, bgr, wbcc
bgr	44	0.731	0.029	age, bu, hemo
sc	17	0.659	0.071	age, bgr, bu
bu	19	0.602	0.070	age, sc, bgr
age	9	0.530	0.069	pcv, hemo, pot

5.1.3 Missingess patterns in Hepatitis Dataset. Distribution of individual missingness rates for the Hepatitis dataset is shown in Table 9. There isn't a single clear drop in the distribution that

would be a natural minimum support threshold. Because of this, we experimented with different values between $[0, 1]$ before choosing 0.06.

Lowering the threshold did not reveal additional meaningful patterns. Instead, it made the ones already found disappear. At 0.05, the `[liver_big, liver_firm]` gets absorbed into a larger itemset that also includes `protime`. Even though the itemset became larger, its any-confidence dropped from 1.00 to 0.80, its all-confidence from 0.91 to 0.12, and its cross-support ratio from 0.91 to 0.15.

We end up with two itemsets: `[liver_big, liver_firm]` with any-confidence 1.000, and `[albumin, alk_phosphate, protime]` with any-confidence 0.625. First represents liver palpation findings, and the second a liver function panel. Those blocks share no variables. The liver palpation block has an all-confidence of 0.909 and a cross-support ratio of 0.909. This means that both variables are almost always missing together. In contrast, the liver function block has an all-confidence of 0.149 and a cross-support ratio of 0.239, so the same cannot be said for this panel. The per-variable conditionals confirm this. `albumin` with conditional 0.625 is more strongly associated with the rest of the block than the remaining two variables (`alk_phosphate` with 0.345 and `protime` with 0.149).

The liver palpation block is missing more frequently in patients who died than in patients who survived (15.6% compared to 4.1%). The liver function block also shows a higher missingness in patients who died (9.4% compared to 5.7%), but this claim isn't as strong.

The liver palpation block missingness rises from 3.2% (2/62) in Grade 1 to 6.0% (4/67) in Grade 2 and 11.1% (1/9) in Grade 3, though the Grade 3 figure rests on a single patient and should be read cautiously. The liver function block shows a different pattern. It is missing only for patients in the Unscorable category and does not appear in Grades 1–3. Since `albumin` is one of the variables required to compute the ALBI score, this association is not clinically informative.

5.1.4 Missingness predictability in Hepatitis Dataset. Table 12 shows the results for hepatitis dataset. The variables in the identified missingness patterns (Subsection 5.1.3) have mean AUC values significantly lower compared to CKD dataset. For pattern that has `liver_big` and `liver_firm`, the mean AUC values are 0.569 and 0.523, respectively. For the variables in liver function block, mean AUC values range from 0.472 for `albumin` to 0.587 for `protime`. The highest mean AUC values are observed for `spleen_palpable` (0.639), `spiders` (0.623) and `ascites` (0.609). These variables each have five missing observations and standard deviations of 0.112, 0.103 and 0.121, respectively. The lowest mean AUC is observed for `albumin` at 0.472.

The feature importance results show that `sgot` and `age` occur frequently in the three highest-ranked predictors for the rest of the variables. For example, `sgot` is in the top three predictors for 9 of 10, and `age` for all 10 variables. The standard deviations varies from 0.034 to 0.139.

Table 12: Predictability of missingness in the Hepatitis dataset.

Variable	Missing	AUC	SD	Top 3 predictors
spleen_palpable	5	0.639	0.112	sgot, age, albumin
spiders	5	0.623	0.103	sgot, age, albumin
ascites	5	0.609	0.121	sgot, age, bilirubin
protime	67	0.587	0.034	age, alk_phosphate, sgot
varices	5	0.580	0.132	sgot, age, albumin
liver_big	10	0.569	0.081	sgot, age, alk_phosphate
alk_phosphate	29	0.555	0.074	sgot, age, albumin
liver_firm	11	0.523	0.090	age, sgot, alk_phosphate
bilirubin	6	0.477	0.139	alk_phosphate, age, protime
albumin	16	0.472	0.061	age, sgot, alk_phosphate

5.2 Synthetic Data Generation With Missingness (RQ2) Results

5.2.1 Downstream predictive performance. Table 13 reports the mean and standard deviation across folds for AUC, F1, sensitivity, specificity, precision, and Brier score for each condition in the CKD and Hepatitis datasets.

For CKD, the three conditions show similar overall performance. The mean AUC is 0.9667 for *R*, 0.9593 for *S1*, and 0.9767 for *S2*, while the corresponding F1 scores are 0.9715, 0.9605, and 0.9774. Sensitivity is 0.9600 for both *R* and *S2*, compared with 0.9320 for *S1*. Specificity and precision are highest under *S2*, at 0.9933 and 0.9961, respectively, while the Brier score is lowest at 0.0275. These differences are relatively small, and the relative performance of the conditions varied across experiments.

For Hepatitis, the differences are more pronounced. AUC is 0.7374 under *R*, compared with 0.6092 under *S1* and 0.6147 under *S2*. F1 follows the same pattern, with mean values of 0.5733, 0.3616, and 0.3820, respectively. Sensitivity is 0.6212 for *R*, compared with 0.3485 for *S1* and 0.3788 for *S2*. Specificity is more similar across conditions, ranging from 0.8293 under *S2* to 0.8699 under *S1*. Precision is 0.5476 for *R*, 0.4097 for *S1*, and 0.4683 for *S2*. The Brier score is lowest under *R* at 0.1942, compared with 0.2388 and 0.2561 for *S1* and *S2*.

The standard deviations indicate greater variability across folds for the Hepatitis dataset for most performance metrics, particularly AUC, F1, sensitivity, and specificity. Given the limited number of folds and the absence of statistical testing, these results are treated as descriptive comparisons.

Overall, the synthetic conditions show relatively similar performance to the real data condition on CKD, but both synthetic conditions show lower performance than *R* for most metrics on Hepatitis. The consistency of these patterns across experiments and their possible causes are considered further in the Subsection 6.2.

5.2.2 Pattern preservation in generated datasets. Table 14 reports the mean, standard deviation, and range of D_{mis} across the 16 draws generated for each branch and dataset.

Table 11: Summary of relevant blocks and their association with outcome and disease severity in the CKD and Hepatitis datasets.

Dataset	Block	Any-conf.	All-conf.	Cross-supp.	Group rates	Outcome/severity association
CKD	Electrolyte panel: [pot, sod]	1.000	0.989	0.989	32.8% CKD 3.3% nonCKD	CKD: G5 6.2% ↑ G4 26.4% ↑ G3b 43.5% ↓ G3a 30.0% ↑ G2 54.5% ↑ G1 55.6% nonCKD: G3a 0.0% ↑ G2 1.6% ↑ G1 2.6%
CKD	Urinalysis: [al, pc, rbc, sg, su]	0.891	0.270	0.303	14.8% CKD 2.7% non-CKD	CKD: G5 21.5% ↓ G4 18.9% ↑ G3b 19.6% ↓ G3a 5.0% ↓ G2 0.0% → G1 0.0% nonCKD: G3a 0.0% → G2 0.0% ↑ G1 3.8%
CKD	Blood count panel: [hemo, pcv, rbcc, wbcc]	0.865	0.344	0.397	16.8% CKD 2.0% non-CKD	CKD: G5 10.8% ↓ G4 9.4% ↑ G3b 19.6% ↑ G3a 20.0% ↑ G2 27.3% ↑ G1 33.3% nonCKD: G3a 0.0% G2 ↑ 1.6% ↑ G1 2.6%
CKD	mp0: complete case	–	–	–	16.8% CKD 74.7% nonCKD	CKD: G5 36.9% ↓ G4 18.9% ↓ G3b 10.9% ↓ G3a 10.0% ↓ G2 4.5% ↓ G1 0.0% nonCKD: G3a 100.0% ↓ G2 80.3% ↓ G1 78.2%
CKD	mp_other: unexplained missingness	–	–	–	31.6% CKD 15.3% nonCKD	CKD: G5 32.3% ↑ G4 35.8% ↓ G3b 28.3% ↑ G3a 45.0% ↓ G2 22.7% ↑ G1 33.3% nonCKD: G3a 0.0%↑ G2 16.4% ↓ G112.8%
Hepatitis	Liver palpation: [liver_big, liver_firm]	1.000	0.909	0.909	15.6% Die 4.1% Live	Die: 3 12.5% → 2 12.5% ↓ 1 0.0% Live: 3 0.0% ↑ 2 3.9% ↓ 1 3.4%
Hepatitis	Liver function: [albumin, alk_phosphate, protime]	0.625	0.149	0.239	9.4% Die 5.7% Live	Unscorable because of ALBI definition
Hepatitis	mp0: complete case	–	–	–	40.6% Die 54.5% Live	Die: 3 37.5% ↑ 2 50.0% ↑ 1 66.7% Live: 3 0.0% ↑ 2 56.9% ↑ 1 64.4%
Hepatitis	mp_other: unexplained missingness	–	–	–	40.6% Die 36.6% Live	Die: 3 50.0% ↓ 2 37.5% ↓ 1 33.3% Live: 3 100.0% ↓ 2 39.2% ↓ 1 32.2%

Table 13: Mean ± standard deviation of performance metrics across folds.

Dataset	Condition	AUC	F1	Sensitivity	Specificity	Precision	Brier
CKD	R	0.9667 ± 0.0263	0.9715 ± 0.0222	0.9600 ± 0.0316	0.9733 ± 0.0279	0.9836 ± 0.0171	0.0350 ± 0.0271
CKD	S1	0.9593 ± 0.0248	0.9605 ± 0.0243	0.9320 ± 0.0415	0.9867 ± 0.0182	0.9915 ± 0.0117	0.0475 ± 0.0285
CKD	S2	0.9767 ± 0.0118	0.9774 ± 0.0137	0.9600 ± 0.0316	0.9933 ± 0.0149	0.9961 ± 0.0088	0.0275 ± 0.0163
Hepatitis	R	0.7374 ± 0.1101	0.5733 ± 0.1617	0.6212 ± 0.1721	0.8536 ± 0.0879	0.5476 ± 0.1860	0.1942 ± 0.0871
Hepatitis	S1	0.6092 ± 0.0501	0.3616 ± 0.0982	0.3485 ± 0.1596	0.8699 ± 0.0614	0.4097 ± 0.0310	0.2388 ± 0.0139
Hepatitis	S2	0.6147 ± 0.0534	0.3820 ± 0.0782	0.3788 ± 0.1144	0.8293 ± 0.1905	0.4683 ± 0.2161	0.2561 ± 0.1349

Table 14: D_{mis} across repeated synthesis draws (mean ± std, range). $n = 16$ per branch per dataset.

Dataset	Branch	D_{mis}	Range
CKD	*S1*	0.218 ± 0.051	[0.150, 0.345]
	S2	0.156 ± 0.037	[0.080, 0.230]
Hepatitis	*S1*	0.102 ± 0.051	[0.026, 0.181]
	S2	0.379 ± 0.063	[0.284, 0.516]

For CKD, *S2* has a lower mean D_{mis} than *S1*, with values of 0.156 and 0.218, respectively. The standard deviation is also lower for *S2* (0.037 compared to 0.051). The ranges overlap, with *S1* ranging from 0.150 to 0.345 and *S2* from 0.080 to 0.230.

For hepatitis, *S1* has a lower mean D_{mis} than *S2*, with values of 0.102 and 0.379, respectively. The standard deviation is 0.051 for *S1* and 0.063 for *S2*. The ranges do not overlap, with *S1* ranging from 0.026 to 0.181 and *S2* from 0.284 to 0.516.

5.3 Subgroup Analysis (RQ3) Results

5.3.1 Subgroup-specific predictive performance. In Table 15 we report Δ_{TPR}, Δ_{TNR}, and the synthetic vs real differences D_{S1},D_{S2} for CKD across 3 independent runs. In Table 16 we report the same for Hepatitis across 5 independent runs.

Table 15: CKD subgroup predictive-performance disparities (mean ± std across 3 runs).

Subgroup	Metric	R	S1	S2	Difference
outlier	Δ_{TPR}	+0.030 ± 0.088	+0.071 ± 0.089	+0.069 ± 0.054	D_{S1} = +0.041 ± 0.080, D_{S2} = +0.039 ± 0.092
	Δ_{TNR}	undefined			
age≥ 65	Δ_{TPR}	+0.007 ± 0.044	−0.039 ± 0.026	−0.028 ± 0.032	D_{S1} = −0.046 ± 0.032, D_{S2} = −0.036 ± 0.068
	Δ_{TNR}	−0.026 ± 0.009	+0.006 ± 0.023	+0.006 ± 0.023	D_{S1} = +0.031 ± 0.029, D_{S2} = +0.031 ± 0.029
htn==yes	Δ_{TPR}	−0.041 ± 0.046	−0.090 ± 0.012	−0.073 ± 0.020	D_{S1} = −0.049 ± 0.034, D_{S2} = −0.032 ± 0.049
	Δ_{TNR}	undefined			
dm==yes	Δ_{TPR}	−0.041 ± 0.024	−0.078 ± 0.051	−0.054 ± 0.021	D_{S1} = −0.037 ± 0.028, D_{S2} = −0.013 ± 0.041
	Δ_{TNR}	undefined			
mp0	Δ_{TPR}	−0.054 ± 0.013	−0.051 ± 0.019	−0.031 ± 0.029	D_{S1} = +0.003 ± 0.019, D_{S2} = +0.023 ± 0.020
	Δ_{TNR}	−0.046 ± 0.029	−0.023 ± 0.010	−0.048 ± 0.033	D_{S1} = +0.023 ± 0.029, D_{S2} = −0.002 ± 0.012

Table 16: Hepatitis subgroup predictive-performance disparities (mean ± std across 5 runs).

Subgroup	Metric	R	S1	S2	Difference
outlier	Δ_{TPR}	−0.101 ± 0.046	−0.221 ± 0.178	−0.191 ± 0.229	D_{S1} = −0.120 ± 0.179, D_{S2} = −0.090 ± 0.265
	Δ_{TNR}	−0.101 ± 0.020	+0.088 ± 0.305	−0.011 ± 0.198	D_{S1} = +0.188 ± 0.293, D_{S2} = +0.090 ± 0.204
age≥ 50	Δ_{TPR}	+0.073 ± 0.137	0.000 ± 0.129	+0.171 ± 0.128	D_{S1} = −0.073 ± 0.212, D_{S2} = +0.098 ± 0.234
	Δ_{TNR}	+0.077 ± 0.083	+0.100 ± 0.108	+0.070 ± 0.071	D_{S1} = +0.022 ± 0.132, D_{S2} = −0.007 ± 0.123
mp0	Δ_{TPR}	−0.249 ± 0.112	+0.052 ± 0.241	−0.049 ± 0.212	D_{S1} = +0.301 ± 0.198, D_{S2} = +0.200 ± 0.240
	Δ_{TNR}	+0.005 ± 0.032	+0.039 ± 0.025	+0.026 ± 0.023	D_{S1} = +0.034 ± 0.026, D_{S2} = +0.021 ± 0.030

For CKD, the results differ between subgroups. For the age≥ 65 subgroup, the TPR disparity changes from +0.007 in the real data to −0.039 for S1 and −0.028 for S2. The TNR disparity changes from −0.026 to +0.005 for both synthetic conditions. For ‘mp0‘, the TPR disparity is similar between the real data and S1, while S2 differs more. The ‘htn==yes‘ and ‘dm==yes‘ subgroups also show changes in TPR disparity, although their TNR disparities are undefined because they contain no non-CKD patients. The ‘outlier‘ subgroup shows a smaller change in TPR disparity, from +0.030 in the real data to +0.071 for S1 and +0.069 for S2. Its TNR disparity is undefined for the same reason.

For Hepatitis, the differences vary across subgroups. For ‘mp0‘, the TPR disparity changes from −0.249 in the real data to +0.052 for S1 and −0.049 for S2. For age≥ 50, it changes from +0.073 to 0.000 for S1 and +0.171 for S2. The ‘outlier‘ subgroup shows a change in the TNR disparity from −0.101 in the real data to +0.088 for S1 and −0.011 for S2. However, the variation across runs is substantial for this subgroup, particularly for S1 (±0.305), so these differences are unreliable.

Overall, the results show that subgroup predictive performance disparities are not reproduced uniformly across the synthetic conditions. The observed differences depend on the dataset and subgroup, and some estimates have substantial variation across runs. This makes it difficult to draw a general conclusion about whether either synthetic condition consistently preserves subgroup predictive performance.

5.3.2 Subgroup-specific privacy exposure. As seen in Table 17, the overall attack rate is higher for CKD than for Hepatitis in both synthetic conditions. For CKD, the ‘outlier‘ subgroup has a higher attack rate than the rest of the population in both *S1* (0.137 vs. 0.063) and *S2* (0.190 vs. 0.062). The ‘htn==yes‘ and ‘dm==yes‘ subgroups also have higher attack rates than the rest of the population, while ‘age≥ 65‘ and ‘mp0‘ do not. For Hepatitis, subgroup attack rates are generally low. The ‘outlier‘ subgroup has no observed attacks in *S1*, but a higher attack rate in *S2* (0.080 vs. 0.005). The other Hepatitis subgroups show only small differences from the rest of the population.

These results suggest that privacy exposure can differ between subgroups, but the pattern is not consistent across datasets or synthetic conditions. In particular, the higher attack rate for the CKD ‘outlier‘ subgroup is observed in both conditions, and the Hepatitis ‘outlier‘ result differs substantially between *S1* and *S2*. The standard deviations show a relatively high variation for some of these estimates, especially the Hepatitis *S2* ‘outlier‘ subgroup (0.080 ± 0.110).

6 DISCUSSION

The results of this thesis suggest that missingness in clinical tabular data can contain structure that is relevant to synthetic data generation and evaluation. However, the results do not show that explicitly representing co-missingness consistently improves the synthetic data. The effect differs between the two datasets and between the evaluation criteria. The subgroup analysis also shows that overall predictive performance and the missingness patterns used in our experiment do not fully describe how synthetic data behaves for different subgroups. In particular, the results suggest that atypical observations may need separate consideration when evaluating privacy, as they can be more exposed to singling-out attacks.

6.1 RQ1: Co-missingness structure

The missingness mining and filtering procedure (Section 4.2) identified a limited number of recognizable co-missingness blocks in both

Table 17: Subgroup singling-out attack rates (mean ± std across 5 runs).

Dataset	Subgroup	Branch	Subgroup rate	Rest rate
CKD	outlier	S1	0.137 ± 0.047	0.063 ± 0.014
	outlier	S2	0.190 ± 0.088	0.062 ± 0.012
	age≥ 65	S1	0.065 ± 0.030	0.067 ± 0.009
	age≥ 65	S2	0.049 ± 0.028	0.074 ± 0.014
	htn==yes	S1	0.099 ± 0.022	0.047 ± 0.009
	htn==yes	S2	0.088 ± 0.014	0.056 ± 0.017
	dm==yes	S1	0.082 ± 0.017	0.059 ± 0.011
	dm==yes	S2	0.079 ± 0.024	0.062 ± 0.011
	mp0	S1	0.063 ± 0.015	0.069 ± 0.013
	mp0	S2	0.051 ± 0.020	0.079 ± 0.014
Hepatitis	outlier	S1	0.000 ± 0.000	0.004 ± 0.006
	outlier	S2	0.080 ± 0.110	0.005 ± 0.009
	age≥ 50	S1	0.005 ± 0.010	0.004 ± 0.005
	age≥ 50	S2	0.019 ± 0.030	0.004 ± 0.005
	mp0	S1	0.003 ± 0.006	0.005 ± 0.007
	mp0	S2	0.008 ± 0.011	0.008 ± 0.012

datasets (Table 11). In CKD, the three relevant blocks correspond to distinct clinical test panels (electrolytes, urinalysis, complete blood count), and their missingness rates differ systematically by CKD status, but especially by disease severity. This is consistent with missingness coming from test ordering decisions, and not from a mechanism unrelated to the patient's condition. It argues against treating missingness in this dataset as MCAR. In Hepatitis, only two blocks were found, and one of them (the liver function panel) shows a severity association that is at least in part an artefact of the ALBI staging definition itself, since albumin is one of the two variables used to compute ALBI. The liver palpation block, by contrast, shows a genuine association with mortality.

The resulting difference between the datasets is also relevant for the interpretation of the subsequent synthesis experiments. CKD shows a more pronounced co-missingness structure under the procedure used here. In this dataset, 35% of patients have at least one of the three identified patterns fully missing, and these patterns account for 55.2% of all missing values. In Hepatitis, only 11.0% of patients have at least one of the two identified patterns fully missing, accounting for 29.9% of all missing values. The identified patterns therefore cover a larger part of both the patient population and the overall missingness in CKD than in Hepatitis.

This difference suggests that the potential value of explicitly representing co-missingness may depend on how clearly missingness patterns can be identified in a dataset. At the same time, the weaker coverage in Hepatitis could also be a result of the mining and filtering procedure used here. The results therefore show that the procedure identified different levels of co-missingness structure in the two datasets, but they do not show that Hepatitis contains no other meaningful patterns that were not retained. This distinction is important when interpreting the subsequent synthesis experiments.

The missingness predictability analysis provides a complementary perspective on the discovered patterns. In CKD (Table 10), the relatively high predictability of missingness for many variables in the identified patterns supports the interpretation that missingness is related to observed clinical characteristics. Variables such as `hemo`, `sc`, and `pcv` are repeatedly among the strongest predictors of missingness, providing some clinical context for these relationships. Their predictive value suggests that whether these measurements are observed is related to other observed characteristics in the dataset. At the same time, predictability is lower for some variables, indicating that the relationship between observed characteristics and missingness is not equally strong across all variables and patterns.

In Hepatitis (Table 12), the weaker predictability provides less evidence that missingness can be inferred from the other observed variables. This is consistent with the weaker co-missingness structure identified by the mining procedure. The two analyses therefore point in the same direction, with both providing stronger evidence of structured missingness in CKD than in Hepatitis. They do, however, describe different aspects of this structure. The mining procedure tries to identify variables that tend to be missing together, and the prediction analysis considers whether the missingness of one variable can be predicted from the observed values of the other variables. Since missingness indicators were excluded from the prediction models, the predictability results provide a separate source of evidence that is inconsistent with the MCAR assumption. However, these results do not allow the underlying missingness mechanism to be classified as MAR or MNAR.

6.2 RQ2: Missingness representation and synthesis

The results for RQ2 do not show a consistent advantage of condition *S2* over condition *S1*. For downstream predictive performance (Table 13), the difference between *S1* and *S2* is relatively small in CKD. Although *S2* is numerically closer to the real data on several metrics in the reported results, these differences are not consistent across runs. In Hepatitis, the two conditions are more similar to each other, but both show a larger reduction in performance compared with CKD. The choice between the two representations therefore does not appear to explain the difference in downstream performance between the datasets. The weaker missingness structure and lower missingness predictability found for Hepatitis in RQ1 provide one possible context for this result. However, the current experiments do not establish that these properties cause the difference in predictive performance.

The missingness-pattern results provide a slightly different picture (Table 14). Here, the better-performing representation changes between the two datasets. In CKD, the co-missingness representation gives a lower D_{mis}, while in Hepatitis it gives a higher value. The repeated draws also show that this difference is not simply due to a comparison of two average values. One possible interpretation is that the joint representation may be more useful when the missingness patterns form clearer groups. This is consistent with the CKD results, where the mining procedure identified more pronounced and clinically recognizable blocks. However, this interpretation should be treated cautiously, since the identified structure also depends on the mining and filtering procedure used in RQ1.

Overall, these results suggest that there is no single missingness representation that is preferable across the two datasets, under

the assumptions for our design. The choice appears to depend on the dataset and on which aspect of synthetic data quality is being evaluated.

6.3 RQ3: Subgroup analysis

For predictive performance, the CKD results (Table 15) show relatively small changes for some subgroups and larger changes for others. Some disparities also change direction between the real and synthetic data. The same general pattern is observed in Hepatitis (Table 16), although some of the differences are larger. Overall, the results do not show a consistent effect of the missingness representation strategy across the different groups. Synthetic data can therefore change the relative performance of a group compared with the remaining population even when the overall predictive performance is similar. The results also differ between groups and metrics, with neither *S1* nor *S2* consistently being closer to the disparity observed in the real data. The standard deviations are also relatively large for several subgroup estimates, especially in Hepatitis. Some of the observed differences therefore vary between synthesis runs and should not be treated as evidence of a real difference.

The privacy results show a somewhat different pattern 17. The CKD outlier group has higher privacy exposure under the synthetic conditions than the remaining patients. This shows that relatively unusual patients can be more exposed to singling-out attacks. However, the analysis does not show why this group is more exposed. The privacy results should also be interpreted separately from the predictive results, since the two evaluations use independently generated synthetic datasets.

The results do not support a general claim that explicitly modeling co-missingness improves subgroup predictive performance or privacy exposure. Instead, they show that these aspects need to be evaluated separately. Good overall predictive performance or good missingness-pattern preservation does not necessarily mean that the synthetic data behaves similarly for all groups or has the same privacy exposure.

7 THREATS TO VALIDITY

7.1 Internal Validity

The synthesis procedure introduces variation between runs because the structure search is not fully deterministic. The pattern preservation analysis uses 16 repeated draws, while the other evaluation results use fewer runs. Some of the differences between *S1* and *S2* may therefore be related to variation between synthesis runs, especially when the differences are small.

The two conditions also differ in more than the type of missingness variable used during synthesis. In *S1*, missingness is reconstructed independently for each variable, while in *S2*, the identified blocks are reconstructed first and residual missingness is then added. The comparison is therefore not a comparison of independent and joint missingness representation alone.

Pattern preservation and downstream predictive performance are evaluated using separate synthesis runs. The results therefore do not show whether the same synthetic dataset that preserves the missingness patterns also gives good downstream predictive performance. The two evaluations describe different aspects of the synthesis strategies.

The clustering, cluster centroids, and outlier thresholds used for the subgroup analysis are determined from the real data before the evaluation and then applied unchanged to the synthetic data. This was necessary to compare the same groups between the real and synthetic data. However, it means that the synthetic data are not used to determine whether the same group structure would be found independently.

7.2 External Validity

The results are based on two clinical tabular datasets with different missingness structures. CKD contains a more pronounced and clinically interpretable co-missingness structure, while the structure identified in Hepatitis is weaker. The results therefore do not show what effects would occur in other clinical datasets with different missingness structures, distributions, or sizes.

Downstream utility is evaluated using the same MICE and logistic regression procedure for all conditions. This keeps the comparison consistent, but the results are specific to this particular downstream setup. A different imputation or prediction method could interact differently with the missingness reconstructed by *S1* and *S2*.

Privacy exposure is evaluated using singling-out attacks generated by Anonymeter. The resulting attack rates therefore depend on this particular attack procedure and do not describe other possible forms of privacy exposure.

The groups used in the subgroup analysis are also specific to the datasets used in this thesis. The predefined groups and the clustering-based outlier group may not represent meaningful groups in other clinical datasets.

7.3 Construct Validity

The minimum support and minimum any-confidence thresholds used to define relevant blocks were selected by testing different threshold values and inspecting the resulting patterns for each dataset. The final choices reflect a judgement about the balance between retaining potentially useful combinations and keeping the number of blocks manageable. Most of this process focused on the CKD dataset, which introduces some bias towards that dataset. Different but equally reasonable choices of support or any-confidence could produce a different set of blocks.

Several choices were also made in the missingness predictability analysis. The minimum of 5 missing or non-missing observations was chosen based on the CKD dataset and then applied to Hepatitis without further adjustment. The datasets differ in size, missingness, and class balance, so this threshold does not represent the same amount of information in both datasets. We also used 20 repetitions as a practical balance between reducing the effect of random imputation and keeping the computational cost manageable. The reported predictability depends on the model and preprocessing used, so different inclusion thresholds, numbers of repetitions, imputation methods, or predictive models could give different AUC values and feature-importance rankings.

The *S2* representation depends on the co-missingness blocks identified by the mining and filtering procedure. These blocks are

mined separately within each training fold, so the patterns available to *S2* depend on the data in that fold. Patterns that are not retained as blocks are not represented jointly and are instead handled using residual missingness rates. The results for *S2* therefore describe the particular patterns identified by this procedure and not joint missingness in the datasets in general.

Missingness outside the identified patterns is reconstructed using residual rates estimated separately for each outcome class. This provides a simple way of adding missingness that is not covered by the mined patterns, but it assumes that these remaining missing values can be added independently. Other dependencies between these missing values are therefore not preserved.

The subgroup analysis also depends on several choices made during subgroup definition. For the clustering-based group, $k = 2$ was selected through a manual comparison of candidate values of k outside the evaluation pipeline. The outlier threshold was defined as two standard deviations above the cluster-specific mean distance to the centroid. Different values of k or different distance thresholds could therefore result in different groups and different outlier groups.

The outlier group is defined using a rule derived from the real data and then applied to the synthetic data. This allows us to compare the same group across conditions, but it does not show whether the same outlier group would be found independently in the synthetic data.

7.4 Conclusion Validity

Some of the predefined and clustering-based groups contain relatively few patients, and some contain only a small number of positive or negative outcomes. Although a TPR or TNR is reported as undefined when an outcome class is completely absent, a metric based on a small number of cases can still change substantially when only a few observations change classification. The subgroup disparities should therefore be interpreted as observed values rather than precise estimates of population-level disparities.

Subgroup predictive performance is evaluated over 3 independent synthesis runs for CKD and 5 for the Hepatitis dataset. These runs capture some of the variation between synthesis runs, but they do not provide a large number of synthetic datasets. The reported means and standard deviations therefore describe the variation observed in these particular runs and may not represent the full variation that would be observed across more synthesis runs.

The privacy analysis has a similar source of uncertainty. Anonymeter's attack-query generation is not seeded, so the subgroup attack rates can vary between repeated runs of the same experiment. Five independent runs are used to reduce the effect of this randomness, but the reported mean and standard deviation still depend on a limited number of attack realizations.

Finally, predictive performance, privacy exposure, and missingness representation are calculated from separately generated synthetic datasets. The results therefore do not show whether a particular level of predictive performance, privacy exposure, and representation occurs at the same time in one synthetic dataset. These results should therefore be interpreted separately rather than as evidence of a joint effect.

8 RELATED WORK

8.1 Synthetic data generation and evaluation

Goncalves et al. [18] did one of the earliest systematic comparisons of synthetic data generation methods for medical tabular data. They evaluated Bayesian networks, mixture models, imputation-based approaches, and GANs across different utility and privacy metrics using cancer registry data. Utility was assessed through statistical fidelity, latent structure similarity, support coverage, and cross-classification. Privacy was assessed through membership and attribute disclosure risks. The main contribution is the cross-classification metric CrCl-SR, which measures if models trained on synthetic data produce results similar to those that were trained on real data. This is an early form of task-specific utility evaluation. Their results showed that no single method dominates across all criteria, and that methods with the highest utility showed the highest disclosure risk, providing empirical evidence for the utility-privacy trade-off. We also evaluate utility and privacy, but focus specifically on whether the way missingness is represented during synthesis changes these results.

Tucker et al. [46] proposed a hybrid Bayesian network for generating high-fidelity synthetic patient data from UK primary care records. They focused on preserving missingness structure, non-linear relationships, and latent variable effects. A key strength of their approach is transparency. By using a probabilistic graphical model, the generative process is interpretable and biases in the original records can be more easily identified and corrected by domain experts. Missingness is modelled through miss states and latent variables. They show that preserving missingness generally produces synthetic data that more closely resembles the original, and argue this is particularly important when missingness is informative. Re-identification risk is evaluated using outlier analysis. Their privacy analysis suggests a low risk of generating identical or near-identical patient records, but they note that patients with rare diseases or unusual demographics are more vulnerable and require more work. Task-specific utility is assessed through sensitivity analyses on machine learning classifiers. The results show that synthetic data has similar distributions and performance to the original data. We also model missingness during synthesis, but instead of using miss states and latent variables, we represent selected relationships between missing variables as discrete patterns and test whether this changes the resulting synthetic data.

Kaabachi et al. [25] did a review of privacy and utility evaluation methods on 73 studies about synthetic health data. They identified 49 different generation methods, 17 utility evaluation methods, but only 5 privacy evaluation methods. Utility was evaluated in 95% of the studies, but only 46% of those motivated by privacy preservation evaluated privacy risks. Most studies relied on similarity-based metrics for both utility and privacy, which the authors show can be mutually conflicting. Low similarity to real data does not guarantee protection against inference attacks. The review also found that fairness evaluation was almost entirely absent, and it appeared in only three cases. The authors proposed a taxonomy that distinguishes between broad utility, narrow utility, fairness, and privacy and called for evaluations that cover all four. We follow this broader view of evaluation by considering downstream predictive performance

together with missingness preservation, subgroup behaviour, and privacy exposure.

Murtaza et al. [32] published a review of 70 studies on synthetic medical data generation. They classified approaches into knowledge-driven, data-driven, and hybrid methods. The main finding is that hybrid methods are the most promising approach that combines interpretability of knowledge models with the efficiency of data supported generation. The review identified two gaps: a lack of standardized evaluation metrics and insufficient attention to longitudinal data. Privacy received considerably less attention than utility or fidelity, with close to half of included studies assuming rather than verifying privacy guarantees. The authors called for a unified quality evaluation framework that covers both utility and privacy. We therefore do not evaluate the synthetic data using a single utility measure, but consider several properties of the generated data.

Snoke et al. [44] make a formal distinction between general and specific utility for synthetic data. General utility measures the similarity of distribution between original and synthetic data. Specific utility measures agreement on the results of a task. The authors used the propensity score mean squared error (pMSE) statistic for the synthetic data and derived its null distribution under correct synthesis. This allows the observed pMSE to be standardized and interpreted relative to what would be expected if the synthesis model were correct. They show that general and specific utility can diverge. In other words, synthesis that performs well on a chosen task may still fail to produce the structure and distribution of the original data. The authors recommend evaluating both dimensions and argue that relying on specific utility alone can give a misleading picture of data quality. We follow this distinction by evaluating missingness pattern preservation as a general property and predictive performance as a task-specific measure of utility.

8.2 Missingness in clinical data

Raman et al. [39] applied the Structural Missing Data Investigations (SMDI) toolkit to describe missingness patterns in a linked EHR-Medicare claims dataset. The toolkit combines three diagnostic tests: comparisons of distributions between observed and missing populations, predictive modelling of missingness, and association testing between missingness indicators and the study outcome. Applied to HbA1c and BMI confounders in a pharmacoepidemiology study, the toolkit identified HbA1c missingness as likely MAR. The strongest predictors were internal medicine visit frequency and BMI missingness, consistent with a clinical pattern where more intensive treatment leads to more regular measurement. The authors also examined co-missingness between the two variables, finding that 22% of patients with any missing value were missing both. The toolkit is designed to make missingness mechanism assessment accessible to non-specialists. We use a similar set of observations about missingness as the starting point for our experiments, but use them to construct missingness representations for synthetic data generation and then test their effect on fidelity and downstream predictive performance.

Tan et al. [45] looked into patterns of missing laboratory data for 232 hospitals in the United States and Europe. They proposed the concept of informative-MNAR (I-MNAR) to describe a very common missingness pattern in clinical data: once a normal laboratory result is obtained, no further tests of the same type appear in the record. The absence of the test is a clinical decision and not a failure to record the data. They show that missing laboratory patterns vary by patient sex, disease severity, and hospital characteristics, and that these patterns can be used to identify relationships between measurements. The key contribution is empirical evidence that missingness in EHR data should not be treated as noise. This provides motivation for treating missingness as part of the data structure.

Wang et al. [49] used a missing pattern as a binary vector to indicate which columns are absent in a given row. They defined the missing pattern distribution as the proportion of each distinct pattern in the dataset. They showed that existing methods that rely on imputation before generation fail to reproduce this distribution in synthetic data. They proposed three GAN-based methods to address this. HottGAN trains a separate generator for every missing pattern. MergeGAN assigns a categorical label to each pattern, adds it as a feature, and trains a generator over the imputed dataset. Empirical evaluation showed that preserving the missing pattern distribution improves synthetic data quality, particularly under MAR and MNAR mechanisms. A key observation is that missing patterns in practice are few and structured. In most real datasets, a few patterns account for over 80% of the samples. Wang et al. treat missing patterns as discrete types to be reproduced rather than as noise that needs to be somehow eliminated. We also represent missingness explicitly during synthesis, but use groups of variables that tend to be missing together rather than treating every complete missingness pattern as a separate category. We then test whether this representation improves preservation of the missingness structure.

8.3 Subgroups, privacy and fairness

Ganev et al. [15] showed the impact of DP on underrepresented subgroups. They tested three DP generative models, namely PrivBayes, DP-WGAN, and PATE-GAN. They varied subgroup sizes and measured accuracy on each under varying privacy budgets and imbalance levels. PrivBayes reduces the gap between majority and minority subgroups, and PATE-GAN increases it. DP-WGAN falls between the two, generally preserving imbalance for moderate privacy budgets but becoming unstable at low epsilon values. Both directions lead to similar impacts on classifier accuracy, with underrepresented subgroups affected more. The magnitude increases with stronger privacy and higher imbalance. The paper also shows that multi-attribute subgroups defined by intersections of age, sex, and race suffer more severe and more variable accuracy drops than single-attribute ones. Additionally, the choice of DP mechanism matters. Different models behave differently under the same privacy budget, and no single one performs better in all settings. We also examine whether synthetic data affects different groups differently, but our analysis does not vary a privacy mechanism or privacy budget. Instead, we compare subgroup predictive performance and singling-out exposure between real and synthetic data.

Pereira et al. [38] proposed an end-to-end evaluation framework for addressing utility and fairness. They don't assume real data is available for evaluation, which makes it useful in settings where

only synthetic data is released. They compared four marginal-based synthesizers (AIM, MWEM PGM, MST, and PrivBayes) with four GAN-based synthesizers using multiple fairness definitions. Marginal methods outperformed GAN methods on both utility and fairness. The key finding is that synthetic data that is good for model training might not be for evaluation. PrivBayes and most GAN-based generators showed this. AIM and MWEM PGM generated data that could support both training and evaluation and preserve fairness for subgroups. The paper also found that performance was independent of the fairness characteristics of the original dataset, which suggests that the choice of algorithm matters more than the properties of the data. We similarly compare predictive performance between groups, but focus on whether the differences observed in real data are preserved after synthesis rather than on comparing synthesizers by their fairness properties.

Bhanot et al. [9] proposed two metrics for fairness: a log disparity metric that measures whether subgroups defined by protected attributes are represented at rates similar to the real data, and a time-series log disparity metric for temporal datasets that measures whether subgroup trends are reproduced. Applied to the MIMIC-III synthetic dataset, the metrics revealed that Black patients and patients who died were highly underrepresented while Whites and Asians were overrepresented. For the temporal datasets, the metrics showed that standard aggregate comparisons can seem fair even though they are masking underrepresentation, in this case for older age groups. A key finding is that synthetic data can satisfy overall utility and resemble the real data while being unfair at the subgroup level. This is why the authors argue that fairness metrics should be an integral part of synthetic data evaluation. We therefore include subgroup-level evaluation and try not to rely only on aggregate similarity. Our analysis differs in that we focus on predictive performance and privacy exposure rather than representation of protected groups.

Kim et al. [26] identified a conflict between privacy and fairness in DP generative models. Naive combinations of DP with fairness reweighting increase sensitivity. That requires more noise to maintain privacy guarantees, which in turn worsens fairness. The opposite direction also fails. Applying DP on top of fair training suppresses minority data usage. They proposed a framework that uses an ensemble of teacher models in two stages to divide fairness from privacy. In the fair training stage, teachers are trained on balanced minibatches. In the private training stage, knowledge is transferred to the generator using differentially private aggregation, with DP guarantees that keep the teacher models private. Experiments on image datasets showed that the framework reduces group size disparity and downstream fairness disparities while preserving comparable utility. We also examine privacy and subgroup behaviour together, but without introducing a differential privacy mechanism. Instead, we use these measures to evaluate whether the synthetic data expose some groups differently from others.

9 CONCLUSION

This thesis investigated how missingness can be represented when generating and evaluating synthetic clinical tabular data. We used the CKD and Hepatitis datasets and found differences in their missingness structure. CKD had more pronounced co-missingness patterns, while the structure identified in Hepatitis was weaker. We also found that the missingness of several variables could be predicted from observed patient characteristics. These results show that missingness contains structure in both datasets, but they do not show whether the underlying missingness mechanism is MAR or MNAR.

We then compared two ways of reconstructing missing values during synthesis. The results differed between the two datasets. In CKD, the approach that explicitly represented the identified co-missingness patterns gave better preservation of the missingness patterns. In Hepatitis, the other approach gave better preservation. The differences in downstream predictive performance were generally smaller. We also found differences between the real and synthetic data for some of the groups used in the evaluation, but neither approach consistently performed better. The privacy analysis showed higher singling-out exposure for the CKD outlier group.

The results therefore do not show a consistent advantage from explicitly representing the selected co-missingness patterns. They also do not show that this approach is generally unhelpful. The results depend on the patterns identified by the mining procedure, the way missingness was reconstructed, the Bayesian network configuration, and the datasets used. The experiments show that the effect of the two approaches differed between CKD and Hepatitis and was not consistently reflected in downstream predictive performance.

9.1 Future Work

One direction for future work is to investigate other ways of representing relationships between missing values. In this thesis, we represent these relationships using discrete patterns found by the mining procedure. We could instead model conditional dependencies between missingness indicators directly. This fits naturally with the Bayesian network model because the network already represents dependencies between variables. We could then check whether the relationships found in the initial analysis are also captured by the network and whether this improves preservation of the missingness structure in the synthetic data.

We could also investigate how the way we fill missing values before synthesis affects the results. In this thesis, we use column-wise imputation so that both synthesis conditions start from the same complete data. This gives us a controlled comparison, but it does not account for different assumptions about why values are missing. We could compare this with imputation methods based on MCAR and MAR assumptions, as well as methods that explicitly account for MNAR. This could show whether the choice of imputation changes the effect of representing relationships between missing values.

The Bayesian network experiments could also use different configurations. The configuration used in this thesis provides a baseline, but it limits some of the ways in which continuous variables can be modelled. We could repeat the experiments using other PyBNesian configurations and compare the resulting synthetic data. We could also compare the Bayesian network with other synthesis models to see whether the same results occur with different models.

The evaluation of predictive performance could also be changed for clinical applications. We use a fixed probability threshold of 0.5 to calculate sensitivity, specificity, precision, and F1. This gives a consistent comparison between the conditions, but a threshold of 0.5 is not necessarily appropriate for a clinical decision. We could instead evaluate these measures across different thresholds or use a threshold based on the intended clinical application.

We could also investigate fairness in more detail. In this thesis, we compare predictive performance and privacy exposure between the groups used in the evaluation, while representation is considered separately. We could use additional fairness measures and examine how representation, predictive performance, and privacy differ between groups. This could give a more complete picture of how the synthetic data affects different groups.

The subgroup definitions could also be changed. We use clustering and an outlier rule to identify groups that may behave differently, but other subgroup discovery methods could produce different results. We could compare different methods and check whether the differences found in this thesis remain when the groups are defined in other ways.

Another direction for future work is to investigate the evaluation metrics in more detail. In this thesis, we use a limited set of metrics to compare the synthetic datasets. Different metrics can measure different aspects of synthetic data quality, so the choice of metrics can affect the conclusions about which missingness representation performs better. Future work could compare more metrics for predictive performance, missingness preservation, subgroup performance, and privacy. This could help determine whether the differences between the two strategies remain when they are evaluated in other ways. It would also allow the experiments to be designed around the evaluation criteria from the start, rather than first changing the synthesis approach and then checking whether the resulting data performs better.

Finally, we could repeat the experiments on larger and more diverse clinical datasets and use more synthesis runs for each condition. Some of the groups in this thesis contain relatively few patients, so percentages can change substantially when only a small number of observations are affected. Larger datasets would give us more observations for these groups. More synthesis runs would also give us a better estimate of the variation between synthetic datasets. However, the current pipeline is computationally expensive, especially when synthesis and evaluation are repeated. Future work should therefore also focus on optimizing the pipeline so that larger datasets and more synthesis runs are practical.

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ETHICS

Based on the USENIX Security Symposium's policy on ethical considerations [2].

A ETHICAL CONSIDERATIONS APPENDIX

Based on the IEEE Security & Privacy Symposium's LLM policy [1].

This thesis uses the CKD and Hepatitis datasets from the UCI Machine Learning Repository. These datasets are publicly available and do not contain directly identifiable patient information. We designed an experiment to measure how two different missingness strategies affect the downstream predictive performance of a Bayesian network model, missingness patterns preservation fidelity, and privacy exposure to singling-out attacks during synthesis. The main stakeholders are researchers working in the field of synthetic data generation, and indirectly, the patients whose data is used in such pipelines during deployment. The intended contribution is methodological. It provides empirical results on trade-offs between fidelity, utility, privacy and fairness across strategies. Our privacy and fairness evaluations as presented in RQ3 are reported as comparative diagnostics. They illustrate where and how much each strategy exposes subgroups to re-identification risk or predictive disparity.

B AI USAGE CONSIDERATIONS APPENDIX

AI tools were used during the research and writing, primarily Claude, ChatGPT, Overleaf's integrated assistant, and Grammarly. Their use included: brainstorming and quick sanity checks when deciding on research directions; as tools to collect material on the background topics; for generating and editing code including parsers for synthetic data, debugging code when it failed, generating tests for code correctness, for Quarto/revealjs visualization code editing; as editorial assistance when writing, including grammar checking and occasional placeholder text to mark directions for further development; occasional summarization when reading source material, and help organizing project files. All AI-assisted inputs, whether code or text, were reviewed and verified by the author before being used in the thesis. AI was not used to design the experimental methodology or to determine the study's conclusions.

ARTIFACTS

Based on the standard Artifact Appendix template used by the USENIX Security Symposium.

C ARTIFACT APPENDIX

C.1 Abstract

The implementation covers the synthetic data evaluation and generation pipeline for CKD and Hepatitis clinical tabular data described in the thesis. It includes missingness pattern mining, two synthesis strategies (*S1*: shadow columns, *S2*: shadow patterns), a real data baseline *R*, and a cross-validated evaluation of predictive performance, fidelity, subgroup fairness, and privacy exposure, as described in the thesis.

C.2 Description & Requirements

C.2.1 Security, privacy, and ethical concerns. None. The artifact only uses the publicly available, de-identified UCI CKD and Hepatitis datasets.

C.2.2 How to access. https://github.com/IvankaPlesica/synth_tab_data/commits/d57244e

C.2.3 Hardware dependencies. None.

C.2.4 Software dependencies. Software dependencies: Python 3.11, with dependencies pinned in requirements.txt (compiled from requirements.in via pip-compile). Includes local build of the semsynth package (dist/semsynth-0.1.0-py3-none-any.whl), developed by Dr. Benno Kruit and used with permission as part of methodology.

C.2.5 Benchmarks. CKD and Hepatitis datasets from the UCI Machine Learning Repository (loaded via pipeline's data/ folder and congif.py dataset registry.

C.3 Set-up

C.3.1 Installation. Clone the repository and install the pinned dependencies inside a virtual environment:

```
git clone https://github.com/IvankaPlesica/synth_tab_data
cd synth_tab_data
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

This installs all required packages, including a local build of the `semsynth` package (`dist/semsynth-0.1.0-py3-none-any.whl`), which is included in the repository.

C.3.2 Basic Test. To verify, run the synthetic data export pipeline on the CKD dataset:

```
python export_synthetic.py --dataset ckd
```

Expected output: the script completes without error and creates the following files in `outputs/`:

- `ckd_synthetic_s1.csv` — *S1* (shadow columns) synthetic data
- `ckd_synthetic_s2.csv` — *S2* (shadow patterns) synthetic data
- `ckd_synthetic_b.csv` — *R*
- `missingness_metadata.json` — mined missingness metadata

No input parameters beyond `–dataset ckd` are required for this basic test. This installation and test have been verified in a clean virtual environment.

C.4 Evaluation workflow

C.4.1 Major Claims.

(C1): *S2* achieves lower D_{mis} than *S1* in the CKD dataset, while *S1* achieves lower D_{mis} than *S2* in the Hepatitis dataset. Reported as comparison (no statistical significance testing) shown in Table 14, reproduced by E1.

(C2): Differences in downstream predictive performance (AUC, F1, sensitivity, specificity, precision, Brier score) between *R*, *S1* and *S2* are relatively small for CKD, and larger but similar

between *S1* and *S2* for Hepatitis. Reported as descriptive comparisons, shown in Table 13, reproduced by E1.

(C3): Subgroup singling-out attack rates show a loose pattern by subgroup type in the CKD dataset (*S1, S2*). The clustering based outlier group and the hypertension/diabetes subgroups tend to show higher exposure than the rest of the population, while the mp0 subgroup and the age≥65 subgroup tend to show similar or lower exposure. In Hepatitis, with less subgroups available, results are not consistent. These are observed tendencies across 5 runs (mean ± std), not statistically tested effects. Shown in Table 17, reproduced by (E2).

(C4): For subgroup predictive performance we show the following: CKD subgroups (age≥65, hypertension, diabetes) show the most consistent pattern. ΔTPR becomes more negative under both *S1* and *S2* relative to *R*. The outlier subgroup shows a similar tendency in both datasets, but the Hepatitis estimates have a big standard deviations. The mp0 subgroup shows a mild tendency toward reduced disparity in CKD but has inconsistent results in Hepatitis. These are exploratory observations across 3 (CKD) or 5 (Hepatitis) runs, not statistically tested effects. Shown in Tables 15,16, reproduced by (E2).

C.4.2 Experiments. Depending on the dataset and included evaluation, execution time varies.

(E1): [25 min]: Run `python evaluate.py –dataset ckd –pipeline-metadata`. Check `outputs/pipeline_metadata.json` for R/S1/S2 performance and D_{mis}, matching RQ2 trends.

(E2): [50 min]: Run `python evaluate.py –dataset ckd –subgroup-metadata –privacy-checks`. Check `outputs/subgroup_metadata.json` for subgroup fairness/privacy results, matching RQ3 trends.

C.5 Version

Based on the LaTeX template for Artifact Evaluation V20231005. Submission, reviewing and badging methodology followed for the evaluation of this artifact can be found at https://secartifacts.github.io/usenixsec2024/.